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Sectional Design Standard for Rigid Organic Printed Boards

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February 1998

A standard developed by IPC

Association Connecting Electronics Industries



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IPC-2222A

Sectional Design Standard for Rigid Organic Printed Boards

Developed by the IPC-2221/2222 Task Group (D-31B) of the Rigid Printed Board Committee (D-30) of IPC

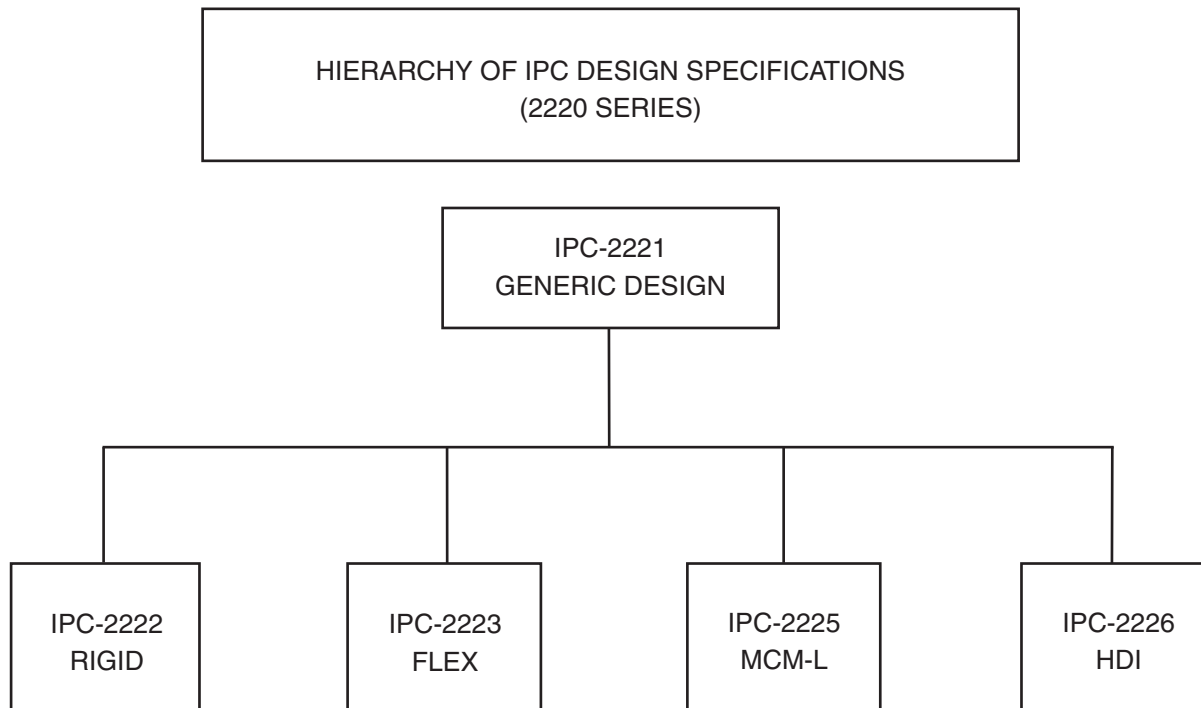
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FOREWORD

This standard is intended to provide information on the detailed requirements for organic rigid printed board design. All aspects and details of the design requirements are addressed to the extent that they can be applied to the unique requirements of those designs that use organic rigid (reinforced) materials or organic materials in combination with inorganic materials (metal, glass, ceramic, etc.) to provide the structure for mounting and interconnecting electronic, electromechanical, and mechanical components.

The information contained herein is intended to supplement generic engineering considerations and design requirements identified in IPC-2221. When coupled with the engineering design input, the complete disclosure should facilitate the appropriate selection process of the materials and the detailed organic rigid structure fabrication technology necessary to meet the engineering design objectives.

The selected component mounting and interconnecting technology for the printed board should be commensurate with the requirements provided and the specific focus of this sectional document.

IPC's documentation strategy is to provide distinct documents that focus on specific aspect of electronic packaging issues. In this regard document sets are used to provide the total information related to a particular electronic packaging topic. A document set is identified by a four digit number that ends in zero (0).

Included in the set is the generic information which is contained in the first document of the set and identified by the four digit set number. The generic standard is supplemented by one or many sectional documents each of which provide specific focus on one aspect of the topic or the technology selected. The designer of the printed board, needs as a minimum, the generic, the sectional of the chosen technology, the generic engineering considerations, and the engineering description of the final product.

Failure to have all information available prior to starting a design may result in a product that is difficult to manufacture or exceeds the cost predictions or expectations of the printed board.

As technology changes, specific focus standards will be updated, or new focus standards added to the document set. The IPC invites input on the effectiveness of the documentation and encourages user response through completion of "Suggestions for Improvement" forms located at the end of each document

Acknowledgment

Any document involving a complex technology draws material from a vast number of sources. While the principal members of the IPC-2221/2222 Task Group (D-31b) of the Rigid Printed Board Committee (D-30) are shown below, it is not possible to include all of those who assisted in the evolution of this standard. To each of them, the members of IPC extend their gratitude.

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Sectional Design Standard for Rigid Organic Printed Boards

1 SCOPE

This standard establishes the specific requirements for the design of rigid organic printed boards.

1.1 Purpose The requirements contained herein are intended to establish specific design details that **shall** be used in conjunction with IPC-2221 to produce designs intended to mount and attach components. The components may be through-hole, surface mount, fine pitch, ultra-fine pitch, array mounting or unpackaged bare die.

The organic materials may be homogeneous, reinforced, or used in combination with inorganic materials; the interconnections may be single, double, or multilayered. They may be any combination able to perform the physical, thermal, environmental, and electronic function.

1.2 Document Hierarchy Document hierarchy **shall** be in accordance with the generic standard IPC-2221.

1.3 Presentation Presentation **shall** be in accordance with the generic standard IPC-2221.

1.4 Interpretation Interpretation **shall** be in accordance with the generic standard IPC-2221.

1.5 Definition of Terms The definition of all terms used herein **shall** be in accordance with IPC-T-50 and as defined in 1.5.1.

1.5.1 As Agreed Between User and Supplier (AABUS) Indicates additional or alternate requirements to be decided between the user and the supplier in the procurement documentation. Examples include contractual requirements, modifications to purchase documentation and information on the drawing. Agreements can be used to define test methods, conditions, frequencies, categories or acceptance criteria within a test, if not already established.

1.6 Classification of Products Classification of products **shall** be in accordance with the generic standard IPC-2221 and as defined in 1.6.1.

1.6.1 Printed Board Type This standard provides design information for different printed board types. Printed board types are classified as:

Type 1 – Single-Sided Printed Board

Type 2 – Double-Sided Printed Board

Type 3 – Multilayer Printed Board without blind or buried vias

Type 4 – Multilayer Printed Board with blind and/or buried vias

Type 5 – Multilayer Metal Core Printed Board without blind or buried vias

Type 6 – Multilayer Metal Core Printed Board with blind and/or buried vias

1.7 Applicability The contents of this standard may not apply to certain leading edge technologies. Refer to IPC-2221 for additional information.

1.8 Revision Level Changes Changes that were incorporated in the current revision of this specification are indicated throughout by gray shading of the relevant subsection(s). Changes to a figure or table are indicated by gray shading of the Figure or Table header.

2 APPLICABLE DOCUMENTS

The following specifications of the revision in effect at the time of order form a part of this document to the extent specified herein. If a conflict of requirements exists between this specification and those listed below, this specification **shall** take precedence.

2.1 IPC¹

IPC-T-50 Terms and Definitions for Interconnecting and Packaging Electronic Circuits

IPC-D-279 Design Guidelines for Reliable Surface Mount Technology Printed Board Assemblies

IPC-TM-650 Test Methods Manual²

2.2.18 Determination of Thickness of Laminates by Mechanical Measurement

2.2.18.1 Determination of Thickness of Metal-Clad Laminates, Cross-sectional

IPC-2152 Standard for Determining Current Carrying Capacity in Printed Board Design

IPC-2221 Generic Standard on Printed Board Design

IPC-4101 Laminate/Prepreg Materials Standard for Printed Boards

IPC-4103 Specification for Base Materials for High Speed/High Frequency Applications

IPC-4411 Specification and Characterization Methods for Nonwoven Para-Aramid Reinforcement

IPC-4562 Metal Foil for Printed Wiring Applications

IPC-6012 Qualification and Performance Specification for Rigid Printed Boards

2.2 Underwriters Laboratories³

UL 746E Standard Polymeric Materials, Materials Used in Printed Wiring Boards

3 GENERAL REQUIREMENTS

General requirements **shall** be in accordance with the generic standards IPC-2221.

3.1 Performance Requirements Finished rigid printed boards **shall** meet the performance requirements of the applicable performance specification (e.g., IPC-6012, IPC-6018, etc.).

4 MATERIALS

4.1 Material Selection Material selection **shall** be in accordance with the generic standard IPC-2221.

4.2 Dielectric Base Materials (Including Prepregs and Adhesives) Dielectric base materials **shall** be in accordance with the generic standard IPC-2221 and the following subsections.

4.2.1 Epoxy Laminates Epoxies are the most common resin materials, which are combined with glass cloth to produce laminates. When compared to other laminate materials, epoxies offer advantages in availability and relative ease in processing. The many different types and blends of epoxies exhibit a wide range of selection for usage or soldering processes; epoxies with

1. www.ipc.org

2. Current and revised test methods are available on the IPC website (www.ipc.org/html/testmethods.htm)

3. www.ul.com

a T_g (glass transition temperature) from 110 °C to in excess of 200 °C are available from most laminate suppliers. Lead-free assembly has become an important material selection criterion. The keyword 'Lead-Free' should be used in IPC-4101 to provide a direct link to suitable materials when lead-free assembly is planned.

4.2.2 High Temperature Laminates High temperature laminates include those made from resins such as Epoxy, Cyanate Ester, Triazine blends and Polyimide. High temperature resin laminates offer the advantages of increased chemical and temperature resistance. Disadvantages include the need for specialized processing and higher material cost. High T_g materials also tend to reduce warpage; however, balanced construction remains the most important factor regarding printed board flatness.

4.2.3 Special Clad Materials The use of surface mount technology may require the use of special clad materials when coefficient of thermal expansion (CTE) matching is critical. Examples of these special materials include copper-clad Invar, epoxy or polyimide with aramid fiber and polyimide/quartz. The most common usage is for Class 3 printed boards, although there may also be some application for Class 2 product. These materials offer the advantages of improved reliability since they can better match component expansion characteristics. Better matched expansion is expected in order to subject solder connections to less stress and provide improved reliability. These materials may require unique processing during printed board fabrication, which is a disadvantage.

4.2.4 Other Laminates Laminates such as paper-based phenolics, etc., have acceptance in some consumer products where the complexity is quite low due to lesser material and manufacturing costs. These materials are associated with very high volume products with lower performance requirements than those usually associated with epoxy type laminates.

4.3 Laminate Materials Laminate materials **shall** be in accordance with the generic standard IPC-2221 and the following paragraphs.

When metal clad, foil type **shall** be as specified in IPC-4562. Unclad laminates without an adhesive, per IPC-4101 or IPC-4103, may be used as fillers in multilayer printed boards for dielectric spacing between layers.

When Underwriter's Labs (UL) requirements are imposed, the material used **shall** be approved by UL for use as fillers in multilayer printed boards for dielectric spacing between layers. Printed boards **shall** be fabricated from the laminate materials specified in Table 4-1 or UL 746E.

The printed board design **shall** be such that temperature rise due to current flow in conductors, when added to all other sources of heat (e.g., heat dissipated by parts mounted on the printed boards, the equipment's general temperature rise, and the specified

Table 4-1 Clad Laminate UL Maximum Operating Temperatures¹

Designation		Dielectric Thickness (min)	Maximum Operating Temperature ^{4, 5, 6, 7}
NEMA	IPC-4101		
FR-4 ²	/21/24/26/27/82/83/ 92/93/94/95/97/98	0.1 mm [0.0040 in]	120 °C
		0.4 mm [0.0157 in]	130 °C
FR-5	23	0.6 mm [0.024 in]	140 °C
		1.4 mm [0.0551 in]	170 °C
GPY ³	30/40/41/42	0.1 mm [0.0040 in]	140 °C
		1.6 mm [0.0630 in]	170 °C
	50	0.1 mm [0.0040 in]	120 °C
		0.4 mm [0.0157 in]	130 °C
	53/60	0.1 mm [0.0040 in]	140 °C
		1.6 mm [0.0630 in]	170 °C
	70/71	0.1 mm [0.0040 in]	140 °C
		0.4 mm [0.0157 in]	170 °C

Note 1. Ambient temperature plus the temperature rise caused by current in the conductors and components.

Note 2. FR-4 laminates should not be combined in one printed board with GPY prepreps.

Note 3. When GPY laminates are combined in one printed board with FR-4 prepreps, the temperature **shall** be that specified for the FR-4 materials.

Note 4. A multilayer printed board **shall** be limited to a maximum operating temperature for the total dielectric thickness shown above.

Note 5. For reinforcement/resin combinations not specifically shown above, the maximum operating temperature should be close to that of a listed combination with the same resin.

Note 6. Other materials in the same classification with higher operating temperatures may be available.

Note 7. Dielectric thickness smaller than those shown above may be temperature rated by UL for certain laminate manufacturers.

operating ambient temperature for the equipment) at the conductor/laminate interface, will not result in an operating temperature in excess of that specified for the laminate material. See IPC-2152 for further discussions on temperature rise in printed boards as a result of current flow in conductors. It is recommended that the designer select material using an appropriate derating factor from the maximum operating temperatures given in Table 4–1. The values in Table 4–1 are based on long term thermal aging tests by UL and may be mandatory for designs to be used in UL approved products. Hot spot temperatures **shall** not exceed the temperatures specified in Table 4–1 for the laminate material selected. Materials used (copper-clad, prepreg, copper foil, heat sink, etc.) **shall** be specified on the master drawing.

Advantages and disadvantages for common laminate materials are given in Table 4–2. This list is not intended to be comprehensive.

Table 4–2 Advantages and Disadvantages of some Common Laminate Materials

Laminate Material (IPC-4101 Designation)	Max. Use Temp., °C	Dielectric Constant (nominal) ³	T _g , °C	CTE (at T < T _g) 10 ⁻⁶ /°C ² x,y z	Finished Printed Board Relative Cost ⁴	Printed Board Producibility ¹	Advantages	Disadvantages
Epoxy/Glass /21 (GF/GFN/GFK)	110	4.5	110 min	15 90	1.0	*****	-Wide Availability -Ease of Processing -Large Printed Board Size	-Poor Reworkability -Max. Temp. 110 °C -Susceptible to thermal induced damage
/23 (GH/GHN/GHP)	170	4.5	135 to 185	15 60	1.0	****	Same as GFN plus: -High temp. use -Reworkability	Limited Availability
/24 (GF/GFG)	130	4.5	150 to 200	15 60	1.0	*****	Same as GFN plus: -High temp. use -Reworkability	-Slightly higher cost than GFN
/25 (GF/GFG)	130	4.5	150 to 200	15 60	2.5	*****	Same as GFN plus: -High temp. use -Reworkability	-Slightly higher cost than GFN
/26 (GFG/GFT)	150	4.5	170 to 220	15 60	1.0	*****	Same as GFG plus: -Higher temp. use -Reworkability	-Slightly higher cost than GFG
Polyimide/Glass /40 (GI/GIN)	170	4.1	200 min	11–15 75	2.5	****	-Higher Temp. Use -Reworkability -Availability	-Higher cost than FR-4 -More Hygroscopic than FR-4
/41 (GI/GIL)	170	4.1	250 min	11–15 75	2.5	****	-Higher Temp. Use -Reworkability -Availability	-Higher cost than FR-4 -More Hygroscopic than FR-4
/42 (GI/GIJ)	170	4.1	200 to 250	11–15 75	2.5	****	-Highest Temp. Use -Reworkability -Availability	-Higher cost than FR-4 -More Hygroscopic than FR-4
Woven Aramid /50 (AF)	125 (up to 140 some matls)	3.6	125 to 190	7–9 100	12.0	**	-Low X-Y CTE -Lightweight -Multifunctional epoxies with T _g of 150–180 °C available	-Max. use temp. 125 °C -Poor Reworkability -Limited Availability
Non-Woven Aramid /53 (BI)	190	3.6	220 min	10–12 110	7.0	***	-Low X-Y CTE -Lightweight -Higher temp & better producibility	-Poor Reworkability -Limited Availability
Cyanate Ester/SGlass /70 (SC)	170	3.5	235	8.5–9.2 44	4.5	**	-Z-axis expansion -Moisture stability -Lightweight	-Limited Availability

Table 4-2 Advantages and Disadvantages of some Common Laminate Materials (cont.)

Laminate Material (IPC-4101 Designation)	Max. Use Temp., °C	Dielectric Constant (nominal) ³	T _g , °C	CTE (at T < T _g) 10 ⁻⁶ /°C ² x,y z	Finished Printed Board Relative Cost ⁴	Printed Board Producibility ¹	Advantages	Disadvantages
Epoxy/Woven Glass /99	150	4.2	180	15-17 55	1.0	*****	-Low Z-axis CTE -High Decomposition Temp	-Higher cost as a lead-free FR-4
Epoxy/Woven Glass /124	150	3.7	180	13-14 65	1.0	*****	-Low Z-axis CTE -High Decomposition Temp	-Higher cost as a lead-free FR-4
Epoxy/Woven Glass /129	150	3.7	180	65	1.0	*****	-Low Z-axis CTE -High Decomposition Temp	-Higher cost as a lead-free FR-4
PTFE IPC-4103	150	2.5 - 10.0	N/A	- -	10.0 -20.0	*	-Low signal loss	-Cold Creep -Poor producibility

Note 1. The greater the number of asterisks given, the better the producibility. Based upon consideration of dimensional stability, material consistency & special processing.

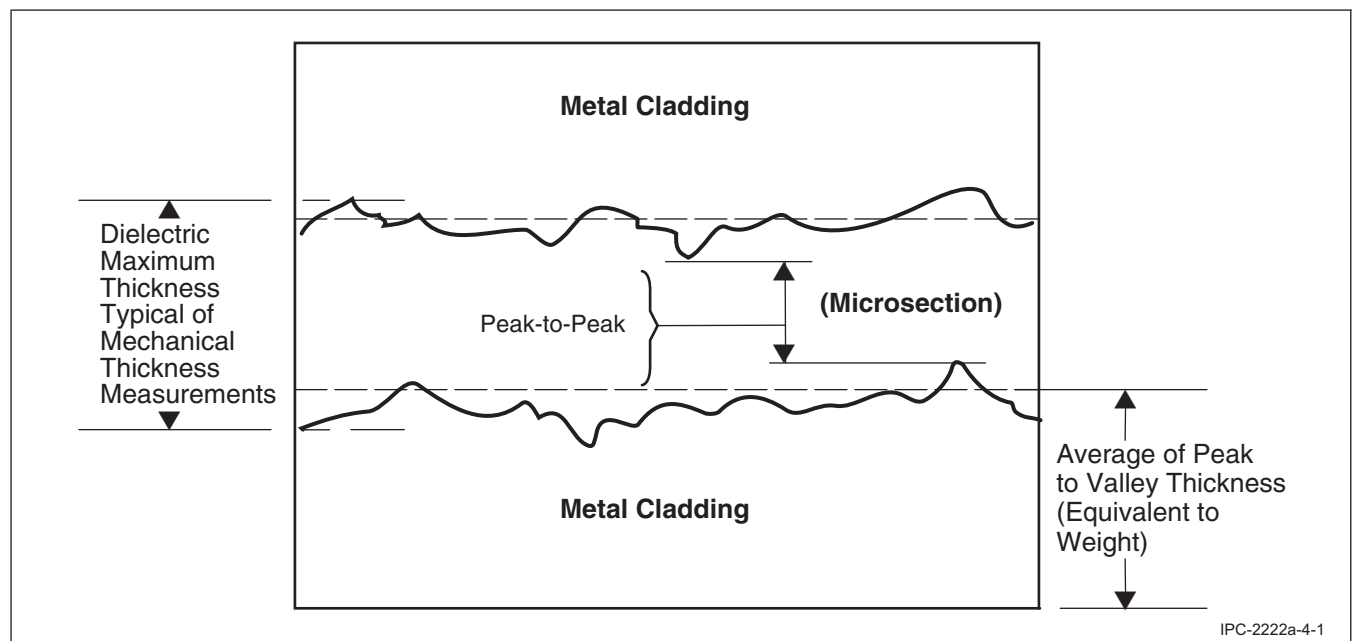
Note 2. For soldering operations, the z-axis CTE above T_g should be considered (will be 2-3 times greater).

Note 3. The dielectric constant values in this table are nominal values as provided in IPC-4101 and IPC-4103. Actual values may be vendor specific.

Note 4. Relative cost factors are in increments of 0.5 with a tolerance of +/- 0.5.

Note 5. The relative cost factor is highly dependent on the material called out.

4.3.1 Measurement of Dielectric Thickness Dielectric thickness will vary across applications. Thickness by mechanical measurement is determined in accordance with IPC-TM-650, Method 2.2.18. Thickness by microsection (as shown in Figure 4-1) is determined in accordance with IPC-TM-650, Method 2.2.18.1. The dielectric thickness is measured in accordance with Figure 4-1 and taken at the closest point between metal claddings.

**Figure 4-1 Dielectric Layer Thickness Measurement**

4.3.2 Dielectric Thickness/Spacing The minimum dielectric thickness/spacing should be specified on the master drawing. If the minimum dielectric spacing and the number of reinforcing layers are not specified, the minimum dielectric spacing defaults to 90 μm [0.00354 in] and the number of reinforcing layers is selected by the supplier. When minimum dielectric spacing is less than 90 μm [0.00354 in], low profile copper foils should be used and the voltages employed should be taken into consideration so as not to cause breakdown between layers. Low profile copper foils may have an adverse effect on availability and cost. When the nominal dielectric thickness on the drawing is less than 90 μm [0.00354 in], the minimum dielectric spacing **shall** be specified on the master drawing, and the number of reinforcing layers may be selected by the supplier. See IPC-2221 for more information on electrical conductor spacing.

If dielectric thicknesses other than the minimum are required by the design, these thicknesses should be clearly defined on the printed board drawing. The copper foil thickness will determine the minimum number of layers of prepreg required to meet the minimum dielectric spacing between conductive layers. For example, the use of 68.60 μm [2 oz./ft²] copper on two adjacent layers will require more sheets of prepreg than if the layers were 34.30 μm [1 oz./ft²] copper, because additional prepreg material is required to fill in the areas between the thicker etched copper patterns.

The base material thickness should be selected from Table 4-3, which lists those thicknesses typically stocked by printed board suppliers. The common copper weights for these thickness laminates are 17.10, 34.30 and 68.60 μm [0.5, 1.0, and 2.0 oz./ft²].

Table 4-3 Common Base Material Thicknesses for Single & Double-Sided Printed Boards, mm [in]

Material Thickness ¹
0.762 [0.030]
1.016 [0.040]
1.57 [0.062]
2.36 [0.093]
3.18 [0.125]

Note 1. Material thickness as measured over laminate (not including copper foil, platings, etc.).

4.3.3 Laminate Properties

4.3.3.1 Laminate Thickness Tolerance When specifying overall multilayer printed board thickness, and individual dielectric thickness between layers, it is important to recognize the effects of accumulated tolerances of individual dielectrics on the overall thickness of completed printed boards.

Laminate material as manufactured has what is known as “working area” as applied to the thickness tolerance. The thickness outside the “working area” of the laminate sheet or cut-to-size panel supplied by the supplier **shall** not vary from the nominal by a value greater than 125% of the specified tolerance. The “working area” default thickness tolerances for printed board laminates are listed in IPC-4101, according to standard laminate thickness range and tolerance classifications. Since only area within the working area is held to the tighter requirements, the manufacturer must fit all of the boards and coupons into this area on the manufacturing panels. Designers should consider this when they define the overall size of their board so that it, and the associated test coupons, will fit in this area.

4.3.3.2 Resin Content Laminates are a composite of resin and glass cloth or other reinforcement. As laminate thicknesses increase, heavier glass cloths or reinforcements are used, and the percentages of resin (resin content) decrease. Laminates with higher resin content tend to have higher coefficients of thermal expansion and lower dimensional stability. Laminates with lower resin content are stiffer; however, if resin contents are too low, weave exposure and measing may result. The glass to resin ratio of a laminate also has a direct effect on dielectric constant.

Unless design constraints dictate, glass style and resin content **shall not** be included on the master drawing, but **shall** only be specified and used in procurement specifications by the board manufacturer.

4.3.4 Prepreg Because of different chemical processing requirements for various resin systems, it is preferred to use like resins for laminates and prepregs when constructing a multilayer printed board.

4.3.4.1 Epoxy Epoxies are the most common resin materials, which are combined with glass cloth or other reinforcement and semi-cured to produce prepregs. Epoxy prepregs will produce acceptable product for all classes of printed boards.

4.3.4.2 High-Temperature Prepregs High temperature prepregs include those made from resins such as Epoxy, Cyanate ester, Triazine blends and Polyimide. High temperature prepregs may be used for specialized Class 2 multilayer printed board applications, but are more commonly used for Class 3 multilayer printed boards.

4.3.4.3 Reinforcement Materials A variety of reinforcement materials, including glass cloth, are available for prepregs and are dependent upon the dielectric thickness and tolerance required, circuit filling needs, and electrical requirements of the dielectric. See IPC-4411.

4.3.4.4 Electrical Requirements For multilayer printed boards, which have controlled impedance requirements, the dielectric constant of the laminated prepreg **shall** be controlled. The designer should indicate controlled impedance requirements on the master drawing. Because dielectric constant is a function of the resin/glass or other reinforcement ratio, prepreg styles should be chosen so that after lamination, the proper retained resin content is achieved in order to arrive at the specified dielectric constant.

4.3.5 Single-Clad Laminates Laminates with foil on one side may be used as an outer layer or internal layer of a multilayer printed board, as appropriate.

4.3.6 Double-Clad Laminates Laminates with foil on both sides may be used to provide either internal or external conductive layers. Double-clad laminate is specified in both industry and IPC specifications by the dielectric separation between the conductive layers as shown in Figure 4–1. Tables 4–4 through 4–8 provide information on the properties of finished bare laminates for different prepreg constructions. To establish final laminate thickness with copper, add 35 μm [0.0014 in] for each oz. of copper on the laminate.

4.3.7 Laminate Material Code Designation Laminate materials **shall** be specified on the drawing. See Figure 4–2 for a mapping of the recommended material selection process.

Materials are generally purchased to meet the requirements of IPC-4101. A typical material code designation of a specific material would be L21 1500 C1/C1 AA (see IPC-4101 for interpretation of material code designations).

4.4 Conductive Materials Conductive materials **shall** be in accordance with the generic standard IPC-2221.

4.5 Organic Protective Coatings Organic protective coatings **shall** be in accordance with the generic standard IPC-2221.

4.6 Markings and Legends Marking and legends **shall** be in accordance with the generic standard IPC-2221.

Slash Sheet 1

REG#	THICKNESS	CONSTRUCTION	% RC	DK	DK. TOL.	DS	Z CTE	THICK TOL	CHEM	MEASLE	AVAIL	COST	FLAT	SMOOTH	DRILL
FR-4 00	0.05mm(0.002")	106	70	4.08	○	○	○	○	●	●	○	○	+	+	+
FR-4 01	0.07mm(0.003")	1080	60	4.25	●	●	○	○	●	●	○	●	+	+	+
FR-4 02	0.08mm(0.003")	2x106	64	4.15	●	○	○	●	●	●	○	○	+	+	+
FR-4 03	0.11mm(0.004")	2x106	72	4.10	●	●	○	●	○	●	○	○	+	+	+
FR-4 04	0.11mm(0.004")	2113	57	4.30	●	○	●	●	○	○	○	●	+	+	+
FR-4 05	0.14mm(0.005")	106/2113	56	4.30	●	●	●	●	●	○	●	○	+	+	+
FR-4 06	0.13mm(0.005")	2x1080	59	4.25	○	⊗	●	○	●	●	●	●	+	+	+
FR-4 07	0.13mm(0.005")	2116	53	4.40	○	○	●	○	○	○	○	○			
FR-4 08	0.16mm(0.006")	106/2116	51	4.45	●	●	○	●	○	○	○	○	—		—
FR-4 09	0.16mm(0.006")	1080/2113	54	4.40	●	○	○	●	○	○	○	○	—		—
FR-4 10	0.18mm(0.007")	2x2113	50	4.50	○	○	○	●	●	○	○	○		+	+
FR-4 11	0.18mm(0.007")	7628	40	4.75	○	○	○	●	⊗	⊗	○	○		—	—
FR-4 12	0.21mm(0.008")	2113/2116	50	4.50	●	●	○	○	○	○	○	○	—		
FR-4 13	0.21mm(0.008")	2x2116	47	4.55	●	○	○	○	○	○	○	○	+	+	+
FR-4 14	0.25mm(0.010")	2x2116	52	4.4	●	○	●	●	●	○	○	○	+	+	+
FR-4 15	0.26mm(0.010")	7628/1080	47	4.6	●	●	○	○	○	○	○	○	—	—	—
FR-4 16	0.26mm(0.010")	2x1080/2116	55	4.6	○	○	○	○	○	○	○	○	+	+	+
FR-4 17	0.31mm(0.012")	2x1080/7628	47	4.6	●	●	○	○	○	○	○	○	+	+	+
FR-4 18	0.32mm(0.012")	7628/2116	47	4.6	●	○	○	○	○	○	○	○	—	—	—
FR-4 19	0.37mm(0.014")	2x7628	41	4.7	●	●	○	○	○	○	○	○	+	+	+
FR-4 20	0.37mm(0.014")	2x2113/7628	46	4.8	○	○	○	○	○	○	○	○	+	+	+
FR-4 21	0.43mm(0.016")	2x2116/7628	48	4.5	○	○	○	○	○	○	○	○			
FR-4 22	0.43mm(0.016")	2x7628/1080	43	4.7	○	○	○	○	○	○	○	○	+	+	+
FR-4 23	0.48mm(0.018")	2x7628/2116	43	4.8	○	○	○	○	○	○	○	○	+	+	+
FR-4 24	0.51mm(0.021")	2x1080/2x7628	46	4.5	○	○	○	○	○	○	○	○	+	+	+
FR-4 25	0.53mm(0.021")	3x7628	40	4.7	○	○	○	○	○	○	○	○	+	+	+
FR-4 26	0.64mm(0.024")	2x2116/2x7628	47	4.6	○	○	○	○	○	○	○	○	+	+	+
FR-4 27	0.61mm(0.024")	3x7628/1080	42	4.7	○	○	○	○	○	○	○	○	—	—	—
FR-4 28	0.74mm(0.028")	4x7628	41	4.7	○	○	○	○	○	○	○	○	+	+	+
FR-4 29	0.74mm(0.028")	2x2113/3x7628	44	4.7	○	○	○	○	○	○	○	○	+	+	+
FR-4 30	0.75mm(0.029")	4x7628/1080	42	4.7	○	○	○	○	○	○	○	○	+	+	+
FR-4 31	1.52mm(0.059")	8x7628	42	4.7	○	○	○	○	○	○	○	○	+	+	+



IPC-2222a-table-4-4

Table 4-4 FR-4 Copper Clad Laminate Construction Selection Guide

Slash Sheet 2

REG#	THICKNESS	CONSTRUCTION	% RC	DK	DK. TOL.	DS	Z CTE	THICK TOL	CHEM	MEASLE	AVAIL	COST	FLAT	SMOOTH	DRILL
HFR-4 00	0.05mm(0.002")	106	70	4.08	○	○	○	○	●	●	○	○	+	+	+
HFR-4 01	0.07mm(0.003")	1080	60	4.25	●	●	○	○	●	●	○	●	+	+	+
HFR-4 02	0.08mm(0.003")	2x106	64	4.15	●	○	○	●	●	●	○	○	+	+	+
HFR-4 03	0.11mm(0.004")	2x106	72	4.10	●	●	○	●	○	●	○	○	+	+	+
HFR-4 04	0.11mm(0.004")	2113	57	4.30	●	○	●	●	○	○	○	●	+	+	+
HFR-4 05	0.14mm(0.005")	106/2113	56	4.30	●	●	●	●	●	○	●	○	+	+	+
HFR-4 06	0.13mm(0.005")	2x1080	59	4.25	○	⊗	●	○	●	●	●	●	+	+	+
HFR-4 07	0.13mm(0.005")	2116	53	4.40	○	○	●	○	○	○	○	○			
HFR-4 08	0.16mm(0.006")	106/2116	51	4.45	●	●	○	●	○	○	○	○	—		—
HFR-4 09	0.16mm(0.006")	1080/2113	54	4.40	●	○	○	●	○	○	○	○	—		—
HFR-4 10	0.18mm(0.007")	2x2113	50	4.50	○	○	○	○	○	○	○	○		+	+
HFR-4 11	0.18mm(0.007")	7628	40	4.75	○	○	○	○	⊗	⊗	○	○		—	—
HFR-4 12	0.21mm(0.008")	2113/2116	50	4.50	●	●	○	○	○	○	○	○	—		—
HFR-4 13	0.21mm(0.008")	2x2116	47	4.55	●	○	○	○	○	○	○	○	+		+
HFR-4 14	0.25mm(0.010")	2x2116	52	4.4	●	○	○	○	○	○	○	○	+		+
HFR-4 15	0.26mm(0.010")	7628/1080	47	4.6	●	○	○	○	○	○	○	○	—	—	—
HFR-4 16	0.26mm(0.010")	2x1080/2116	55	4.6	○	○	○	○	○	○	○	○	+	+	+
HFR-4 17	0.31mm(0.012")	2x1080/7628	47	4.6	●	○	○	○	○	○	○	○	+	+	+
HFR-4 18	0.32mm(0.012")	7628/2116	47	4.6	●	○	○	○	○	○	○	○	—	—	—
HFR-4 19	0.37mm(0.014")	2x7628	41	4.7	●	○	○	○	○	○	○	○	+	+	+
HFR-4 20	0.37mm(0.014")	2x2113/7628	46	4.8	○	○	○	○	○	○	○	○	+	+	+
HFR-4 21	0.43mm(0.016")	2x2116/7628	48	4.5	○	○	○	○	○	○	○	○			
HFR-4 22	0.43mm(0.016")	2x7628/1080	43	4.7	○	○	○	○	○	○	○	○	+	+	+
HFR-4 23	0.48mm(0.018")	2x7628/2116	43	4.8	○	○	○	○	○	○	○	○	+	+	+
HFR-4 24	0.51mm(0.021")	2x1080/2x7628	46	4.5	○	○	○	○	○	○	○	○	+	+	+
HFR-4 25	0.53mm(0.021")	3x7628	40	4.7	○	○	○	○	○	○	○	○	+	+	+
HFR-4 26	0.64mm(0.024")	2x2116/2x7628	47	4.6	○	○	○	○	○	○	○	○	+	+	+
HFR-4 27	0.61mm(0.024")	3x7628/1080	42	4.7	○	○	○	○	○	○	○	○	—	—	—
HFR-4 28	0.74mm(0.028")	4x7628	41	4.7	○	○	○	○	○	○	○	○	+	+	+
HFR-4 29	0.74mm(0.028")	2x2113/3x7628	44	4.7	○	○	○	○	○	○	○	○	+	+	+
HFR-4 30	0.75mm(0.029")	4x7628/1080	42	4.7	○	○	○	○	○	○	○	○	+	+	+
HFR-4 31	1.52mm(0.059")	8x7628	42	4.7	○	○	○	○	○	○	○	○	+	+	+



IPC-2222a-table-4-5

Table 4-5 High T_G FR-4 Copper Clad Laminate Construction Selection Guide

Slash Sheet 3 **CYANATE ESTER (170 to 230 Tg) COPPER CLAD LAMINATE CONSTRUCTION SELECTION GUIDE**

REG#	THICKNESS	CONSTRUCTION	% RC	DK	DK. TOL.	DS	Z CTE	THICK TOL	CHEM	MEASLE	AVAIL	COST	FLAT	SMOOTH	DRILL
CE 00	0.05mm(0.002")	106	70	3.10	○	○	○	○	●	●	○	○	○	+	+
CE 01	0.07mm(0.003")	1080	62	3.37	●	●	○	○	●	●	○	●	+	+	+
CE 02	0.08mm(0.003")	2x106	60	3.44	●	○	○	○	●	●	○	○	+	+	+
CE 03	0.11mm(0.004")	2x106	68	3.28	●	●	○	○	○	○	○	○	+	+	+
CE 04	0.11mm(0.004")	2113	54	3.70	●	○	○	○	○	○	○	○	+	+	+
CE 05	0.14mm(0.005")	106/2113	52	3.71	●	●	○	○	○	○	○	○	+	+	+
CE 06	0.13mm(0.005")	2x1080	56	3.58	○	○	○	○	○	○	○	○	+	+	+
CE 07	0.13mm(0.005")	2116	52	3.71	○	○	○	○	○	○	○	○	+	+	+
CE 08	0.16mm(0.006")	106/2116	51	3.80	●	●	○	○	○	○	○	○	+	+	+
CE 09	0.16mm(0.006")	1080/2113	51	3.80	●	○	○	○	○	○	○	○	+	+	+
CE 10	0.18mm(0.007")	2x2113	50	3.78	○	○	○	○	○	○	○	○	+	+	+
CE 11	0.18mm(0.007")	7628	40	4.08	○	○	○	○	○	○	○	○	+	+	+
CE 12	0.21mm(0.008")	2113/2116	48	3.85	○	○	○	○	○	○	○	○	+	+	+
CE 13	0.21mm(0.008")	2x2116	44	3.90	○	○	○	○	○	○	○	○	+	+	+
CE 14	0.25mm(0.010")	2x2116	52	3.71	○	○	○	○	○	○	○	○	+	+	+
CE 15	0.26mm(0.010")	7628/1080	45	3.93	○	○	○	○	○	○	○	○	+	+	+
CE 16	0.26mm(0.010")	2x1080/2116	54	3.70	○	○	○	○	○	○	○	○	+	+	+
CE 17	0.31mm(0.012")	2x1080/7628	45	3.93	○	○	○	○	○	○	○	○	+	+	+
CE 18	0.32mm(0.012")	7628/2116	44	3.90	○	○	○	○	○	○	○	○	+	+	+
CE 19	0.37mm(0.014")	2x7628	38	4.15	○	○	○	○	○	○	○	○	+	+	+
CE 20	0.37mm(0.014")	2x2113/7628	44	3.90	○	○	○	○	○	○	○	○	+	+	+
CE 21	0.43mm(0.016")	2x2116/7628	44	3.90	○	○	○	○	○	○	○	○	+	+	+
CE 22	0.43mm(0.016")	2x7628/1080	39	4.11	○	○	○	○	○	○	○	○	+	+	+
CE 23	0.48mm(0.018")	2x7628/2116	39	4.11	○	○	○	○	○	○	○	○	+	+	+
CE 24	0.51mm(0.021")	2x1080/2x7628	41	4.05	○	○	○	○	○	○	○	○	+	+	+
CE 25	0.53mm(0.021")	3x7628	38	4.15	○	○	○	○	○	○	○	○	+	+	+
CE 26	0.64mm(0.024")	2x2116/2x7628	44	3.90	○	○	○	○	○	○	○	○	+	+	+
CE 27	0.61mm(0.024")	3x7628/1080	41	4.05	○	○	○	○	○	○	○	○	+	+	+
CE 28	0.74mm(0.028")	4x7628	38	4.15	○	○	○	○	○	○	○	○	+	+	+
CE 29	0.74mm(0.028")	2x2113/3x7628	41	4.05	○	○	○	○	○	○	○	○	+	+	+
CE 30	0.75mm(0.029")	4x7628/1080	38	4.15	○	○	○	○	○	○	○	○	+	+	+
CE 31	1.52mm(0.059")	8x7628	40	4.08	○	○	○	○	○	○	○	○	+	+	+



IPC-2222a-table-4-6

Table 4-6 Cyanate Ester (170 to 250° T_g) Copper Clad Laminate Construction Selection Guide

Slash Sheet 4

REG#	THICKNESS	CONSTRUCTION	% RC	DK	DK. TOL.	DS	Z CTE	THICK TOL	CHEM	MEASLE	AVAIL	COST	FLAT	SMOOTH	DRILL
BT 00	0.05mm(0.002")	106	70	3.60	○	○	○	○	●	●	○	○	○	+	+
BT 01	0.07mm(0.003")	1080	60	3.73	●	●	○	○	●	●	○	●	+	+	+
BT 02	0.08mm(0.003")	2x106	64	3.65	●	○	○	●	●	●	○	○	+	+	+
BT 03	0.11mm(0.004")	2x106	72	3.55	●	●	○	●	○	○	○	○	+	+	+
BT 04	0.11mm(0.004")	2113	57	3.81	●	○	●	●	○	○	○	○	+	+	+
BT 05	0.14mm(0.005")	106/2113	56	3.83	●	●	○	●	○	○	○	○	+	+	+
BT 06	0.13mm(0.005")	2x1080	59	3.75	○	○	○	○	○	○	○	○	+	+	+
BT 07	0.13mm(0.005")	2116	53	3.92	○	○	○	○	○	○	○	○	+	+	+
BT 08	0.16mm(0.006")	106/2116	51	3.97	●	●	○	●	○	○	○	○	+	+	+
BT 09	0.16mm(0.006")	1080/2113	54	3.89	●	○	○	○	○	○	○	○	+	+	+
BT 10	0.18mm(0.007")	2x2113	50	4.00	○	○	○	○	○	○	○	○	+	+	+
BT 11	0.18mm(0.007")	7628	40	4.30	○	○	○	○	○	○	○	○	+	+	+
BT 12	0.21mm(0.008")	2113/2116	50	4.00	○	○	○	○	○	○	○	○	+	+	+
BT 13	0.21mm(0.008")	2x2116	47	4.10	○	○	○	○	○	○	○	○	+	+	+
BT 14	0.25mm(0.010")	2x2116	52	4.0	○	○	○	○	○	○	○	○	+	+	+
BT 15	0.26mm(0.010")	7628/1080	47	4.1	○	○	○	○	○	○	○	○	+	+	+
BT 16	0.26mm(0.010")	2x1080/2116	55	3.9	○	○	○	○	○	○	○	○	+	+	+
BT 17	0.31mm(0.012")	2x1080/7628	47	4.1	○	○	○	○	○	○	○	○	+	+	+
BT 18	0.32mm(0.012")	7628/2116	47	4.1	○	○	○	○	○	○	○	○	+	+	+
BT 19	0.37mm(0.014")	2x7628	41	4.3	○	○	○	○	○	○	○	○	+	+	+
BT 20	0.37mm(0.014")	2x2113/7628	46	4.2	○	○	○	○	○	○	○	○	+	+	+
BT 21	0.43mm(0.016")	2x2116/7628	48	4.1	○	○	○	○	○	○	○	○	+	+	+
BT 22	0.43mm(0.016")	2x7628/1080	43	4.3	○	○	○	○	○	○	○	○	+	+	+
BT 23	0.48mm(0.018")	2x7628/2116	43	4.3	○	○	○	○	○	○	○	○	+	+	+
BT 24	0.51mm(0.021")	2x1080/2x7628	46	4.2	○	○	○	○	○	○	○	○	+	+	+
BT 25	0.53mm(0.021")	3x7628	40	4.3	○	○	○	○	○	○	○	○	+	+	+
BT 26	0.64mm(0.024")	2x2116/2x7628	47	4.1	○	○	○	○	○	○	○	○	+	+	+
BT 27	0.61mm(0.024")	3x7628/1080	42	4.3	○	○	○	○	○	○	○	○	+	+	+
BT 28	0.74mm(0.028")	4x7628	41	4.3	○	○	○	○	○	○	○	○	+	+	+
BT 29	0.74mm(0.028")	2x2113/3x7628	44	4.2	○	○	○	○	○	○	○	○	+	+	+
BT 30	0.75mm(0.029")	4x7628/1080	42	4.3	○	○	○	○	○	○	○	○	+	+	+
BT 31	1.52mm(0.059")	8x7628	42	4.3	○	○	○	○	○	○	○	○	+	+	+



IPC-2222a-table-4-7

Table 4-7 BT Copper Clad Laminate Construction Selection Guide

Slash Sheet 5

POLYIMIDE COPPER CLAD LAMINATE CONSTRUCTION SELECTION GUIDE

REG#	THICKNESS	CONSTRUCTION	% RC	DK	DK. TOL.	DS	Z CTE	THICK TOL	CHEM	MEASLE	AVAIL	COST	FLAT	SMOOTH	DRILL
POLY 00	0.05mm(0.002")	106	72	4.00	○	○	○	○	●	●	○	●	+	+	+
POLY 01	0.07mm(0.003")	1080	63	4.02	●	●	○	○	●	●	○	●	+	+	+
POLY 02	0.08mm(0.003")	2x106	61	4.05	●	○	○	○	●	●	○	○	+	+	+
POLY 03	0.11mm(0.004")	2x106	59	3.89	●	●	○	○	○	●	○	○	+	+	+
POLY 04	0.11mm(0.004")	2113	55	4.21	●	○	○	○	○	○	○	○	+	+	+
POLY 05	0.14mm(0.005")	106/2113	53	4.26	●	●	○	○	○	○	○	○	+	+	+
POLY 06	0.13mm(0.005")	2x1080	57	4.15	○	○	○	○	○	○	○	○	+	+	+
POLY 07	0.13mm(0.005")	2116	54	4.23	○	○	○	○	○	○	○	○	+	+	+
POLY 08	0.16mm(0.006")	106/2116	52	4.28	●	●	○	○	○	○	○	○	+	+	+
POLY 09	0.16mm(0.006")	1080/2113	52	4.28	●	○	○	○	○	○	○	○	+	+	+
POLY 10	0.18mm(0.007")	2x2113	50	4.34	○	○	○	○	○	○	○	○	+	+	+
POLY 11	0.18mm(0.007")	7628	39	4.66	○	○	○	○	○	○	○	○	+	+	+
POLY 12	0.21mm(0.008")	2113/2116	49	4.37	○	○	○	○	○	○	○	○	+	+	+
POLY 13	0.21mm(0.008")	2x2116	45	4.48	○	○	○	○	○	○	○	○	+	+	+
POLY 14	0.25mm(0.010")	2x2116	54	4.23	○	○	○	○	○	○	○	○	+	+	+
POLY 15	0.26mm(0.010")	7628/1080	42	4.42	○	○	○	○	○	○	○	○	+	+	+
POLY 16	0.26mm(0.010")	2x1080/2116	56	4.18	○	○	○	○	○	○	○	○	+	+	+
POLY 17	0.31mm(0.012")	2x1080/7628	47	4.42	○	○	○	○	○	○	○	○	+	+	+
POLY 18	0.32mm(0.012")	7628/2116	46	4.45	○	○	○	○	○	○	○	○	+	+	+
POLY 19	0.37mm(0.014")	2x7628	39	4.66	○	○	○	○	○	○	○	○	+	+	+
POLY 20	0.37mm(0.014")	2x2113/7628	45	4.48	○	○	○	○	○	○	○	○	+	+	+
POLY 21	0.43mm(0.016")	2x2116/7628	45	4.18	○	○	○	○	○	○	○	○	+	+	+
POLY 22	0.43mm(0.016")	2x7628/1080	41	4.60	○	○	○	○	○	○	○	○	+	+	+
POLY 23	0.48mm(0.018")	2x7628/2116	40	4.63	○	○	○	○	○	○	○	○	+	+	+
POLY 24	0.51mm(0.021")	2x1080/2x7628	42	4.57	○	○	○	○	○	○	○	○	+	+	+
POLY 25	0.53mm(0.021")	3x7628	39	4.66	○	○	○	○	○	○	○	○	+	+	+
POLY 26	0.64mm(0.024")	2x2116/2x7628	45	4.48	○	○	○	○	○	○	○	○	+	+	+
POLY 27	0.61mm(0.024")	3x7628/1080	42	4.57	○	○	○	○	○	○	○	○	+	+	+
POLY 28	0.74mm(0.028")	4x7628	39	4.66	○	○	○	○	○	○	○	○	+	+	+
POLY 29	0.74mm(0.028")	2x2113/3x7628	42	4.57	○	○	○	○	○	○	○	○	+	+	+
POLY 30	0.75mm(0.029")	4x7628/1080	40	4.63	○	○	○	○	○	○	○	○	+	+	+
POLY 31	1.52mm(0.059")	8x7628	41	4.60	○	○	○	○	○	○	○	○	+	+	+



IPC-2222a-table-4-8

Table 4-8 Polyimide Copper Clad Laminate Construction Selection Guide

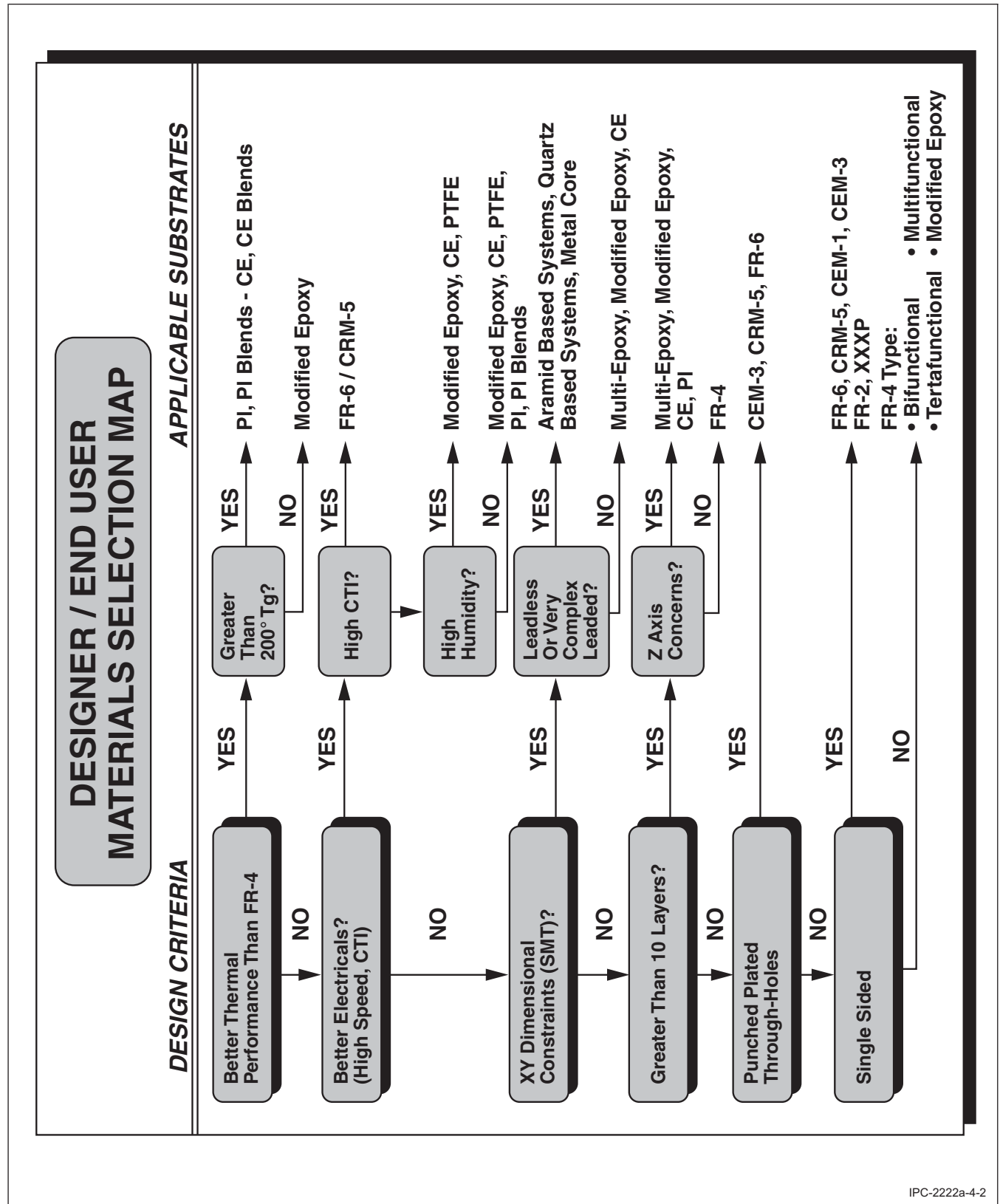


Figure 4-2 Designer/End User Materials Selection Map

5 MECHANICAL/PHYSICAL PROPERTIES

5.1 Fabrication Requirements Fabrication requirements **shall** be in accordance with the generic standard IPC-2221 and as follows:

Typical printed board processing equipment is sized to accept the industry standard, 457 x 610 mm [18.0 x 24.0 in] panels. The designer should consult with printed board manufacturers if the design will require larger panel sizes. The manufacturing base for such designs will be limited.

5.2 Product/Printed Board Configuration Product/ printed board configuration **shall** be in accordance with the generic standard IPC-2221 and as follows:

5.2.1 Printed Board Geometries The following subsections provide considerations to be taken into account during the design of a printed board.

5.2.1.1 Borders and Spacing Borders and margins are commonly employed by the printed board fabricator to provide room for tooling features and other process control features (see Figure 5–1).

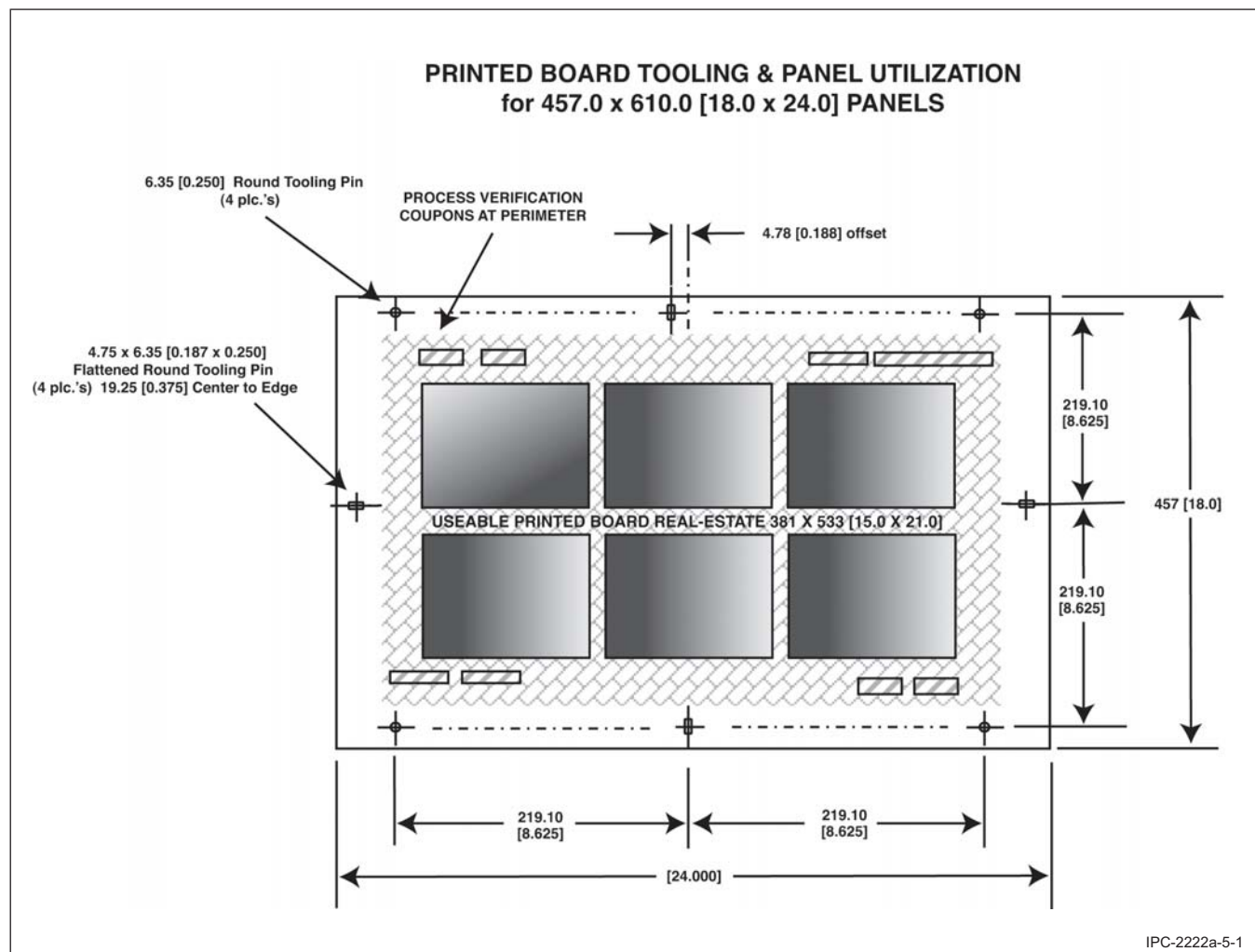


Figure 5–1 Borders and Margins in Printed Board Panels, mm [in]

The size of such borders is usually in the range of 19.1 to 40.0 mm [0.75 to 1.575 in]. This is determined by taking into account an optimum number of printed boards per panel, obtaining optimum plating across the panel (which is especially important for high-density/fine line printed boards), etc. Test coupons, when used, are located within these borders.

When the printed boards are made using a print-and-etch procedure, the border size may depend on the type of printed board being made, i.e., double-sided printed boards tend to have smaller borders, while multilayer printed boards tend to have wider borders. The size of the borders need not be the same on all four sides of the panel.

The separations between printed boards or between test coupons and printed boards on a panel have to accommodate the panel/printed board shearing, blanking and routing operations. Typical board to board separation is 6.35 mm [0.25 in], but may range from 2.54 mm [0.10 in] to 12.7 mm [0.50 in].

5.2.1.2 Dimensional Aspect Ratio Printed board length to width relationships should be kept as similar as possible. Long narrow printed boards or unusually shaped printed boards lead to excessive bowing/twisting. Dimensional instability and associated problems at all stages of fabrication, assembly, test and system fixturing become factors in determining final printed board size.

5.2.2 Mechanical Support Adequate mechanical support should be provided typically for at least two opposite edges of a printed board assembly. The location and method of support **shall** be such as to minimize shock and/or vibration to a level that will protect against fracturing or loosening of conductor foil, or breaking of the components or component leads as a result of flexing the printed board assembly within the tolerance of the applicable specification.

5.3 Design Requirements for Printed Board Assembly Design requirements for printed board assembly **shall** be in accordance with the generic standard IPC-2221 and as follows in 5.3.1.

5.3.1 Assembly, Palletization and Test Palletization of parts is a standard process used in many instances for both test and assembly. This can be achieved using a number of different techniques. These include simple scoring, a combination of routing and scoring, and a combination of routing plus breakaway.

Small printed boards can effectively be arranged on a single working panel, if the designer works closely with manufacturing. Panel construction may include several printed boards arranged in a matrix or simply one printed board requiring additional material retained for efficient assembly processing. The large printed board or several smaller printed boards are retained in the panels and separated after all assembly processes are completed. Excising or separating the individual printed boards from the panel **shall** be planned as well. Several methods are used to retain circuits in a panel, including V-groove scoring, routed slot with break-away tabs and mouse bites.

Scoring is the machining of a shallow, precise V-groove into the top and bottom surfaces of the laminate in a straight line. It is generally accomplished using CNC equipment. As scoring allows the removal of rails and individual parts from a pallet, positional accuracy is critical. See Table 5–1 and Figure 5–2 for some standard scoring parameters.

Note: A small cross section of printed board material is retained at the break line. An allowance for the scoring angle **shall** be made as well. Conductors that are located too close to the score groove will be exposed or damaged, and rough edges **shall** be sanded lightly to remove burrs and rough fabric particles. See Figure 5–3.

Routing is the process of profiling a pallet or printed board to the correct dimension using a cutting bit. This can be performed either by the use of a pin router and template, or a CNC routing machine.

Frequently a combination of routing and scoring is used, where both pallet and printed board are routed, leaving a small connection bridge. This bridge is then scored to facilitate removal following test and assembly.

The final method involves the use of routing and drilled breakaway tabs. Instead of scoring, a series of holes are drilled in the tab to facilitate removal (See Figure 5–4).

The routed slot and tab pattern is widely used for panel construction and break-away tab extensions. Routing is more precise than scoring, and edge surfaces are smooth, but the break-away “tab” points will require consideration. Tabs can be cut and ground flush with the printed board edge or pre-drilled in a pattern. The drilled pattern furnishes a low stress break point on the “tab”. If the hole pattern is recessed within the printed board edge, secondary sanding or grinding can be bypassed. See Figure 5–5 and Figure 5–6.

Table 5–1 Standard Scoring Parameters, μm [in]

Detail Letter	Title	Definition	Attainable Tolerances
A	Web	The material remaining between the two (2) 'V' scores on a plane perpendicular to the printed board surface	± 80 [0.00315]
B	Centrality	The distance the center of a web is offset from true center within the printed board	± 80 [0.00315]
C	Blade offset	The distance the top and bottom scoring blades are offset from one another	± 80 [0.00315]
D	Score width	The width of a score line at the surface of the printed board	± 80 [0.00315]
E	Cutter angle	The total angle of a scoring blade	$\pm 2^\circ$
F	Keep out area	The area, expressed from nominal score line placement, that no features should be placed within	D/2 + All registration
G	Printed board thickness	Overall printed board thickness to be scored	See 5.5
H	Trueness/Position	The tolerance of two or more score lines on one side of the printed board. Measured from nominal, squareness and actual position.	± 80 [0.00315] cumulative

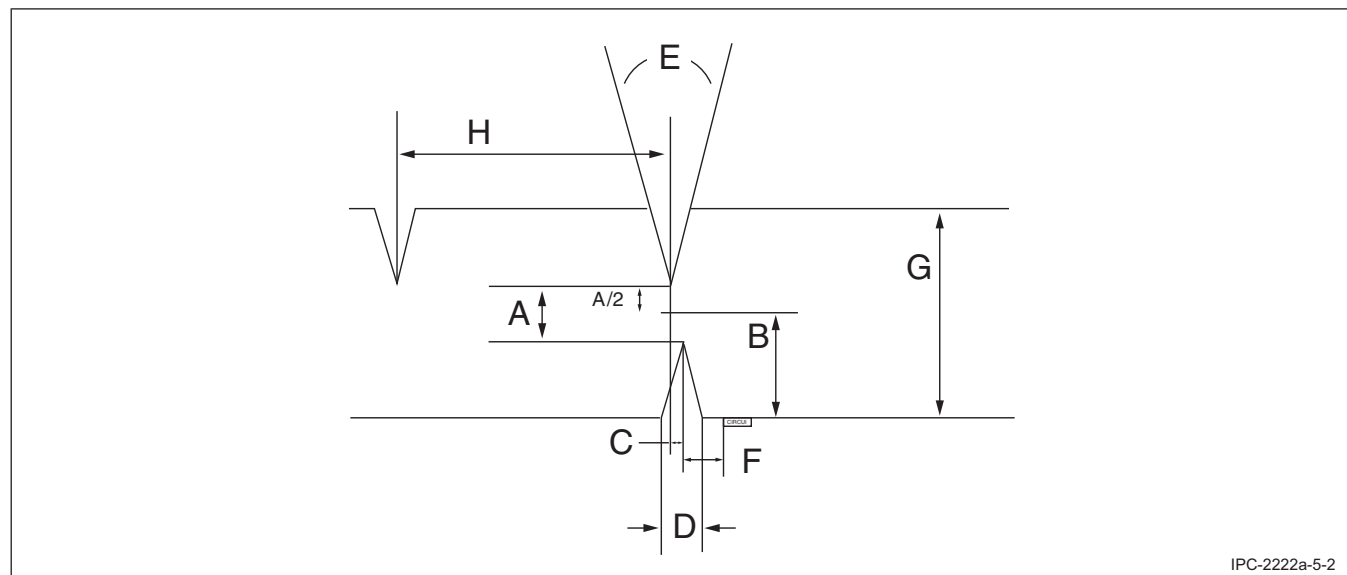


Figure 5–2 Scoring Parameters

Design consideration should be given to the alignment of the mouse bites as shown in Figure 5–6. If the center of the line of holes is even with the edge of the printed board, after the tabs are snapped, the edge of the printed board will be “scalloped” in that location. If the edge of the hole is even with the edge of the printed board, the scalloped area will be outside the printed board edge.

All methods used to create break-aways to allow printed boards to be removed from assembly pallets have the potential to break earlier than intended. This can cause significant assembly problems and/or scrap.

5.4 Dimensioning Systems Dimensioning systems, including grid systems, **shall** be in accordance with the generic standard IPC-2221 and as follows.

5.4.1 Profiles, Cutouts and Notches It is recommended that the number of cutouts and notches on the printed board be kept to a minimum in order to decrease the amount of time and effort necessary to fabricate them. This ultimately will help to minimize end product printed board cost.

All such cutouts and notches **shall** not interfere with the fabrication of other printed board/assembly features, such as the plating of edge-printed board contacts. The edges of internal cutouts **shall** be considered as external printed board edges and meet all

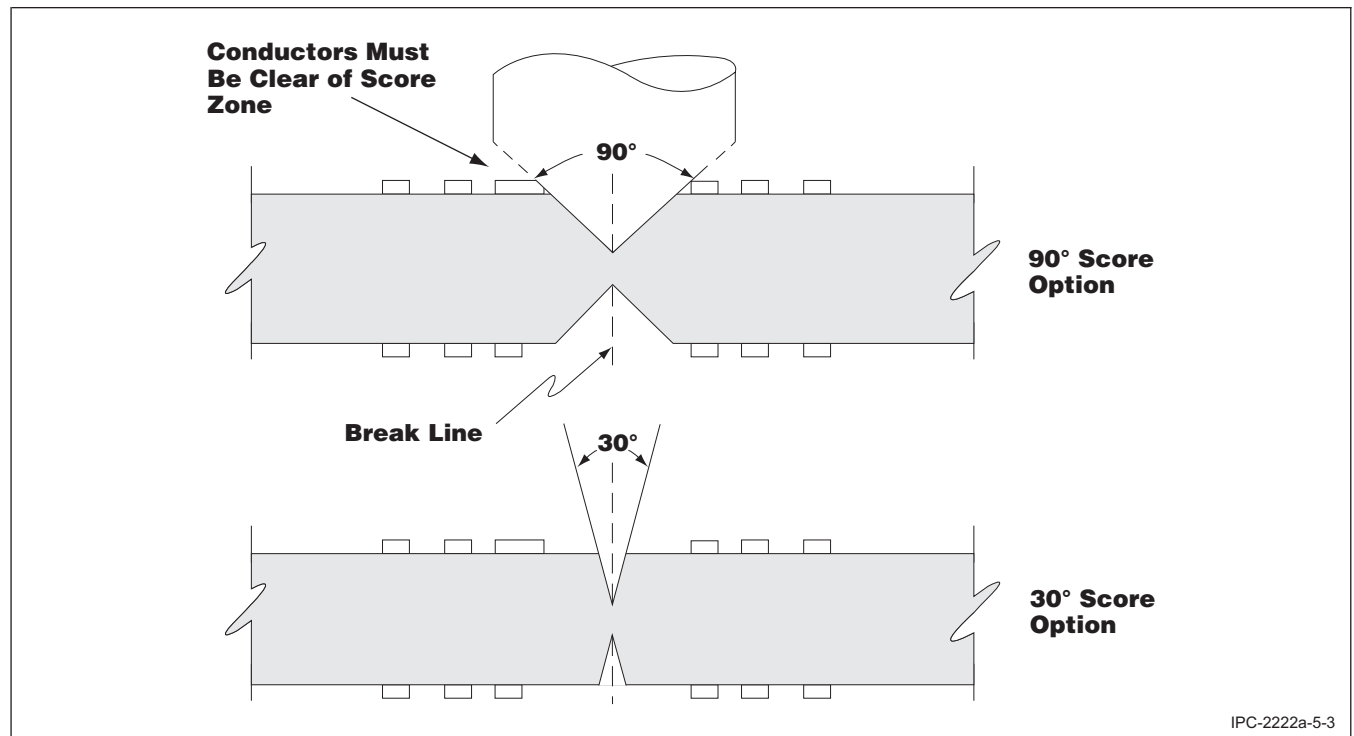


Figure 5-3 Conductor Clearance for V-Groove Scoring

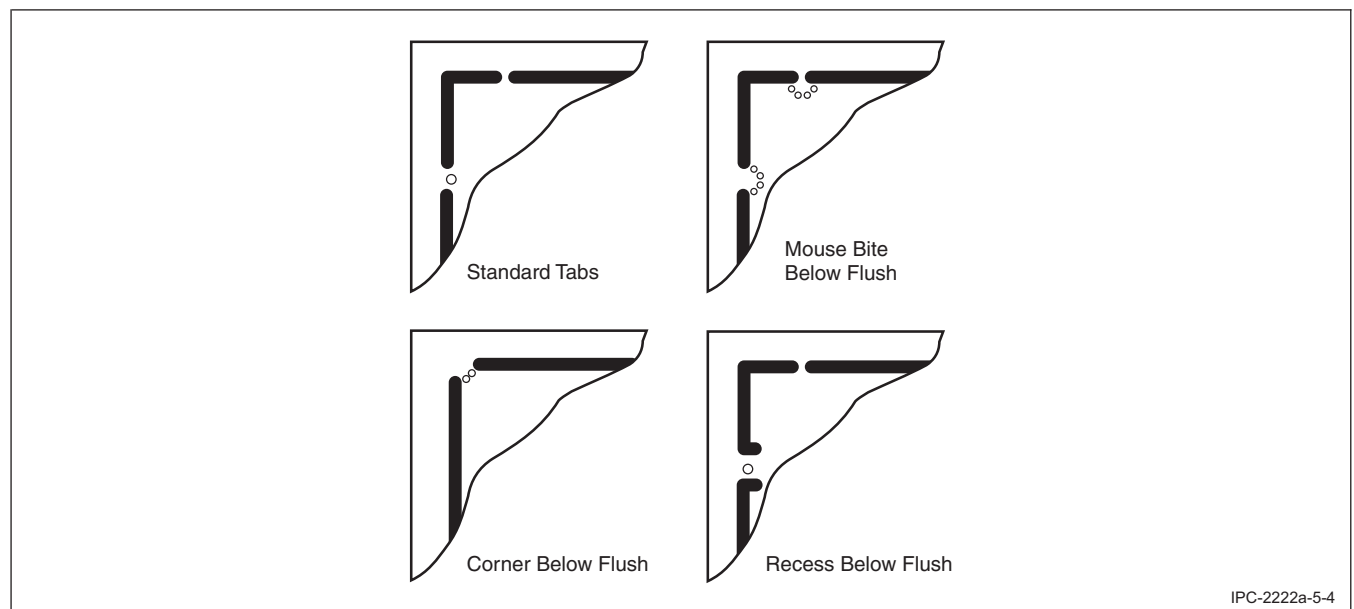


Figure 5-4 Breakaway Tabs

requirements for hole and conductor to edge clearances. On multilayer printed boards the location of lands, clearances, ground planes and other conductive pattern features relative to cutouts and notches **shall** be evaluated for appropriate clearance and tolerance buildup (see 10.1.1).

For printed board shape routing, it is recommended that the cutouts and notches allow for a minimum of a 1.5 mm [0.059 in] radius on internal corners (assuming that a 3.0 mm [0.118 in] diameter router bit is used). Although a minimum internal radius of 0.3mm [0.012 in] can be obtained with a 0.6mm [0.024 in] diameter router bit, smaller internal corner radius requirements than 1.5 mm [0.059 in] or 1.2 mm [0.047 in] should be avoided as router efficiency and accuracy decreases substantially when smaller router bits are used. Recommended tolerances for the location and profile of cutouts and notches are shown in Table 5-2; however, the tolerances specified on the printed board drawing **shall** accommodate the dimensions and tolerances of the mating part.

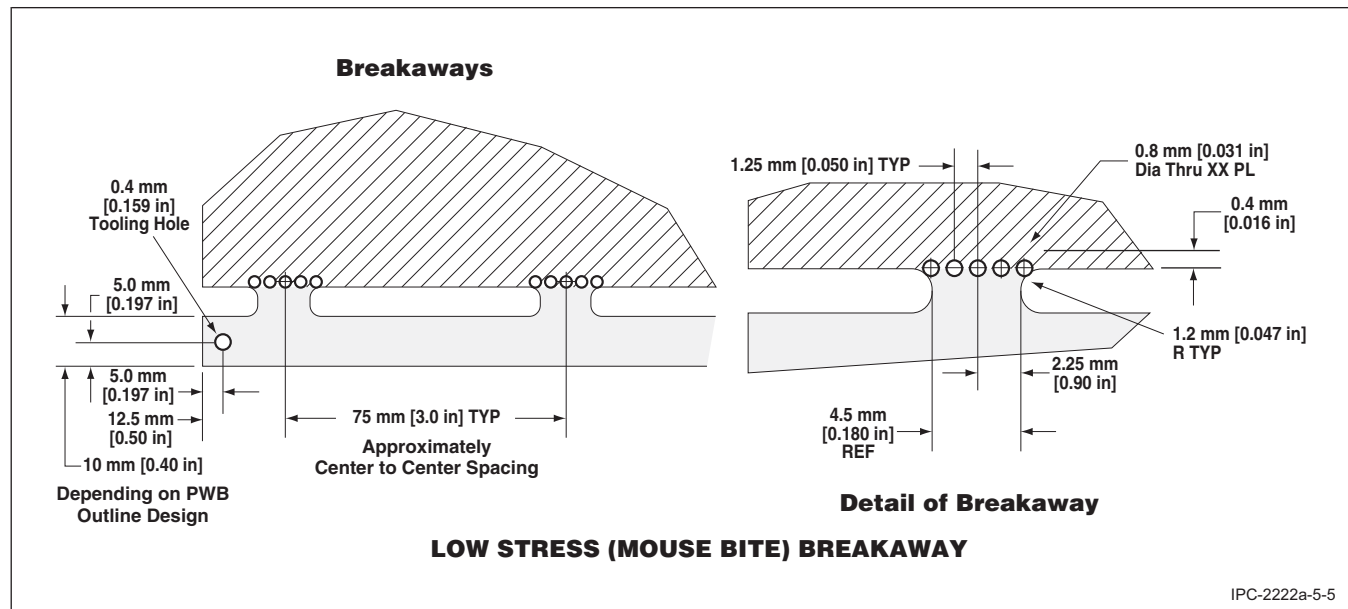


Figure 5-5 Breakaway (Routed Pattern) with Routed Slots

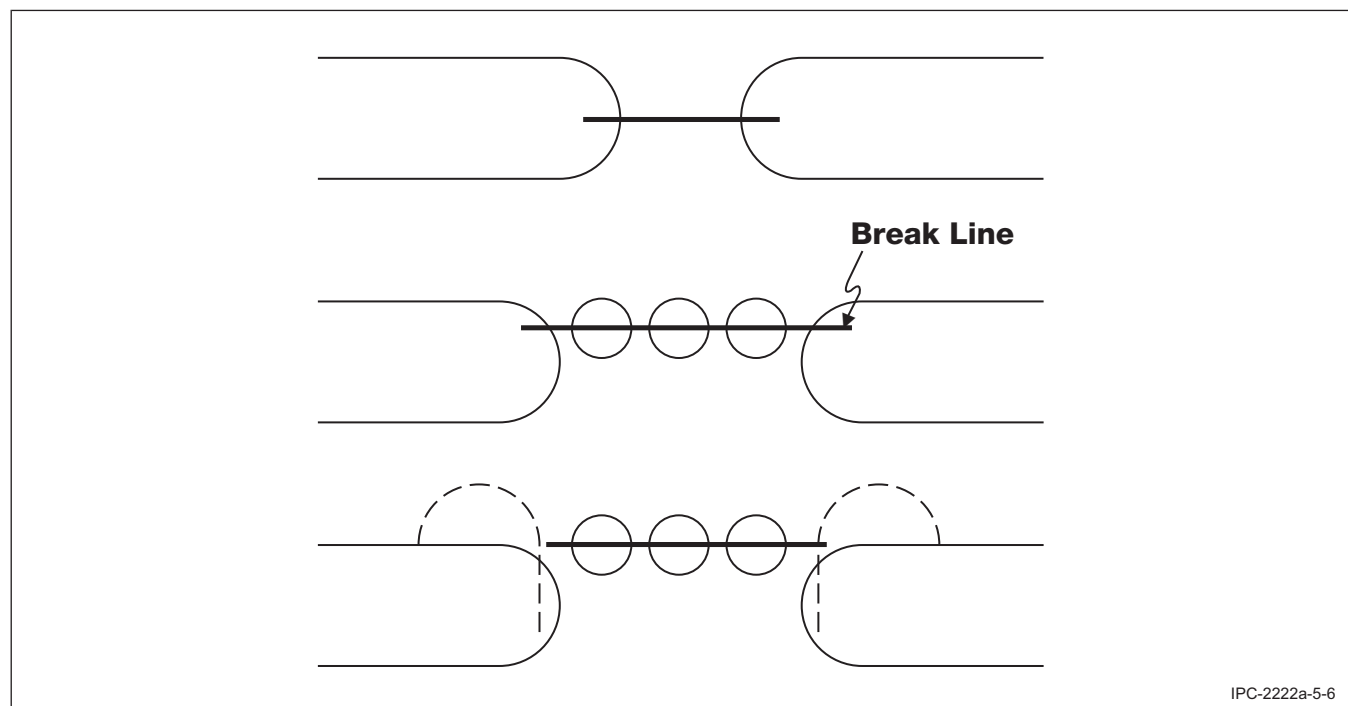


Figure 5-6 Routed Slots

Table 5-2 Tolerance of Profiles, Cutouts, Notches, and Keying Slots, as Machined, mm [in]

Tolerances to be applied to profile of a surface	Level A ¹	Level B	Level C
Profile feature	0.25 [0.00984]	0.20 [0.00787]	0.15 [0.00591]
Location where greatest basic location dimension is less than 300.0 [11.8]	0.30 [0.0118]	0.25 [0.00984]	0.20 [0.00787]
Location where greatest basic location dimension is greater than 300.0 [11.8]	0.35 [0.0138]	0.30 [0.0118]	0.25 [0.00984]

Note 1. For definition of producibility level, see IPC-2221.

5.5 Printed Board Thickness Tolerance On multilayer printed boards, when determining the finished thickness of the printed board (the total, overall laminated or pressed thickness, including external layer foil/cladding and platings), the individual pressed dielectric thickness between layers (base laminate and prepreg), the accumulated tolerances of the individual base laminate and prepreg starter materials, copper foils, platings, and the nominal resin flow or scaled flow thickness of the prepregs, should be considered in the specification of the overall thickness of the finished printed board. The specification of materials used on printed boards requiring tightly toleranced thicknesses should involve the consultation and concurrence of the printed board manufacturer. See Table 5-3 for how the accumulated thickness tolerances apply to the overall printed board thickness tolerance. The values in Table 5-3 are based on laminate to laminate measurements. If measuring over finishes or coatings, then those thicknesses and tolerances need to be considered.

Table 5-3 Printed Board Thickness Tolerance Levels

Level A	Level B	Level C
+/-15% or +/- [0.010 in], whichever is greater	+/-10% or +/- [0.007 in], whichever is greater	+/- 7% or +/- [0.005 in], whichever is greater

6 ELECTRICAL PROPERTIES

Electrical properties **shall** be in accordance with the generic standard IPC-2221.

7 THERMAL MANAGEMENT

Thermal management **shall** be in accordance with the generic standard IPC-2221.

8 COMPONENT AND ASSEMBLY ISSUES

Component assembly issues **shall** be in accordance with the generic standard IPC-2221 and as follows:

8.1 General Attachment Requirements In addition to the general attachment requirements outlined in the generic standard IPC-2221, the following **shall** apply:

8.1.1 Attachment of Wires/Leads to Terminals For cases in which more than one wire is attached to a terminal, the largest diameter wire should be mounted to the bottom-most post for ease of removal and repair. In general, no more than three attachments should be made to each section of a turret or bifurcated terminal. As an exception, bus bar terminals may hold more than three wires or leads per section when specifically designed to hold more.

8.1.2 Printed Board Extractors Printed board extractors or handles are used to provide a convenient means of extracting the printed board from its mating connector. They are generally used where the amount of force makes it difficult to safely remove the printed board without damage to the electrical components or to the person removing the printed board.

Printed board extractors are commercially available and come in a variety of shapes and sizes.

Extractors are usually of the camming type and are mounted to the corners of the printed board. They provide a mechanical advantage for disengaging the connectors and a convenient place to grasp the printed board during removal.

Printed board extractors may be incorporated into the design of the printed board, or may require separate conditions in the printed board assembly. When printed board extractors are a part of the design, adequate reinforcement **shall** be used to properly allow the extracting action to remove the printed board from its connected assembly in the backplane (see Figure 8-1).

When printed board extractors are not a part of the printed board assembly, an extractor of the gripping variety may be used (see Figure 8-2). They grip the printed board in a particular area, which **shall** be kept free of components and circuitry. If a hook-type printed board extractor is used, where a hook passes through holes in the printed board, and then pulls the printed board out, special grommets should be used to reinforce the hole structure to avoid printed board crazing or cracking.

Metallic printed board extractors **shall** have an adequate protective finish to avoid corrosion. Cam type extractors are mounted to the corners of the printed board assembly where they provide a mechanical advantage for disengaging the printed board assembly from the mating connector. They also provide a convenient place to grasp the printed board assembly during removal.

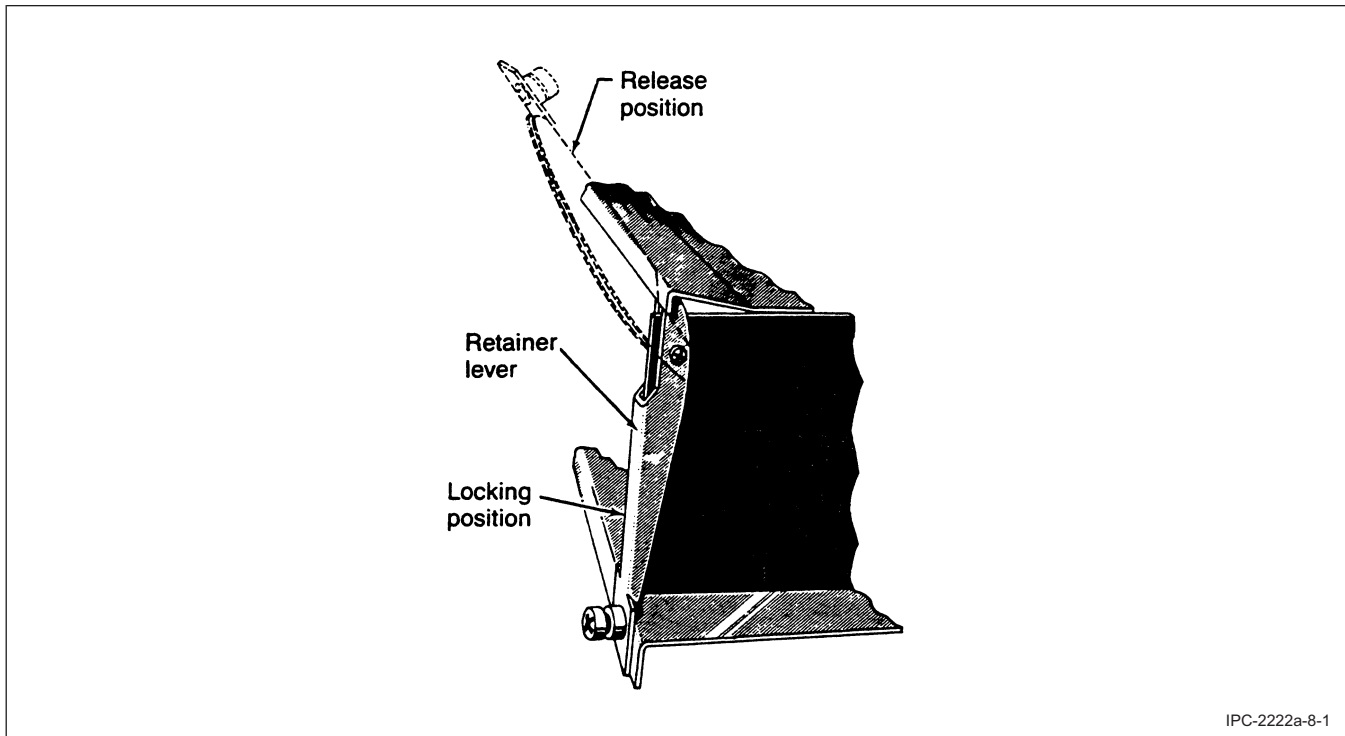


Figure 8-1 Permanent Printed Board Extractor

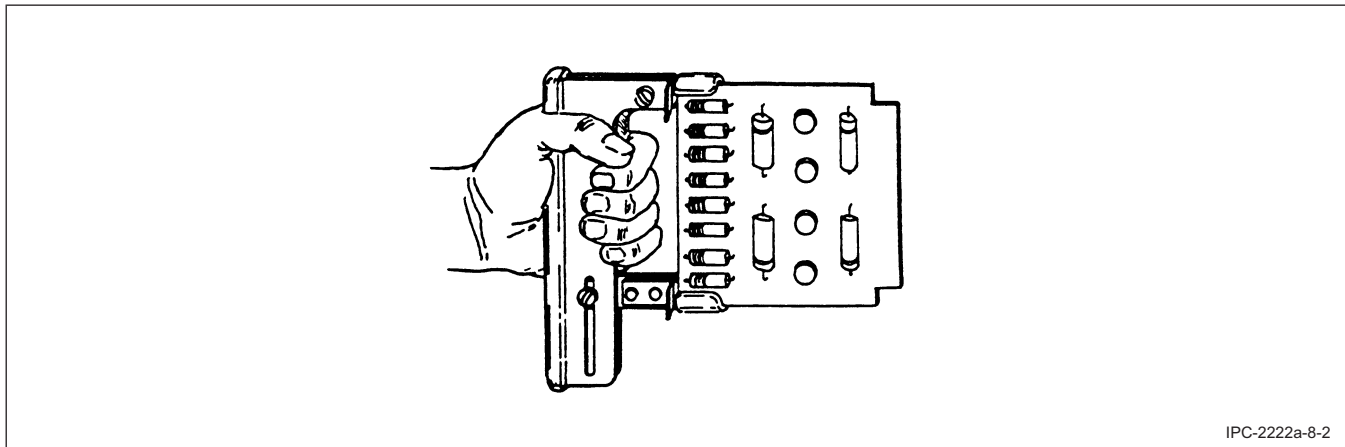


Figure 8-2 External Printed Board Extractor

9 HOLE/INTERCONNECTIONS

9.1 General Requirements for Lands with Holes General requirements for lands with holes **shall** be in accordance with the generic standard IPC-2221 and as follows:

9.1.1 Land Requirements When eyelets or standoff terminals are used, the lands on external layers **shall** be so designed as to have a minimum diameter of at least 0.5 mm [0.0197 in] greater than the maximum diameter of the projection of the eyelet of solder terminal flange.

9.1.2 Thermal Relief in Conductor Planes The relationship between the hole size, land and web area is critical in order to provide successful thermal relief during printed board assembly. The total thermal tie cross-sectional area including connections on multiple planes **shall** meet the minimum current carrying capacity requirements for a given net. If the individual web width violates the intended minimum conductor width it **shall** be specified on the master drawing.

Typically, divide 60% of the minimum land area diameter by the number of webs desired to obtain the width of each web in accordance with the following example:

A. Land Size Calculation, mm [in]

Maximum hole size = 1.0 [0.0394]

Annular ring = 2×0.05 [0.00197] = 0.10 [0.00394]

Fabrication allowance = 0.25 [0.00984 in]

Minimum land size = $1.0 + 0.10 + 0.25$ [0.0394 + 0.00394 + 0.00984] = 1.35 [0.0531] diameter

B. Thermal Relief Calculation, mm [in]

Total thermal width = 60% of land size = 0.6×1.35 [0.0531] = 0.80 [0.0315 in]

C. Original Web Size Calculation, mm [in]

2-web width = 1/2 of total thermal width = 0.50×0.80 [0.0197 x 0.0315] = 0.40 [0.0157]

3-web width = 1/3 of total thermal width = 0.33×0.80 [0.0130 x 0.0315] = 0.27 [0.0106]

4-web width = 1/4 of total thermal width = 0.25×0.80 [0.00984 x 0.0315] = 0.20 [0.00789]

If the actual land diameter chosen is greater than the minimum value calculated, the percentage difference between the land diameters **shall** be subtracted from the total web width calculation, i.e.:

Minimum land diameter = 1.35 [0.0531]

Actual land diameter = 1.70 [0.0670]

Percent difference = $(1.70 - 1.35) / 1.35$ [(0.0670 - 0.0531) / 0.0531] = 25%

New total web width = total web width percent difference = $0.80 \text{ mm} - 25\% (0.80 \text{ mm})$ [0.0315 - 25% (0.0315)] = 0.60 [0.0236]

D. Adjusted Web Size Calculation, mm [in]

2-web width = 1/2 of new total web width = 0.50×0.60 [0.0197 x 0.0236] = 0.30 [0.0118]

3-web width = 1/3 of new total web width = 0.33×0.60 [0.0130 x 0.0236] = 0.20 [0.00789]

4-web width = 1/4 of new total web width = 0.25×0.60 [0.00984 x 0.0236] = 0.15 [0.0060]

Since heat will dissipate through all connections within the pad stack, the total cumulative copper web for all layers in any PTH should not exceed 4.0 mm [0.157 in] for 34.80 μm [1 oz.] copper or 2.0 mm [0.0787 in] for 68.60 μm [2 oz.] copper.

9.1.3 Clearance Area in Planes Clearance areas in planes **shall** be based on the electrical clearance, the drill size, the manufacturing allowance and the land size as follows (see Figure 9–1):

Clearance size associated with a land (functional or nonfunctional) is:

Clearance diameter = minimum land size in accordance with IPC-2221 + (2* pad clearance in Table 9–1)

Clearance size associated with a plated hole (without a land) is:

Clearance diameter = minimum land size in accordance with IPC-2221 – (2* minimum annular ring) + (2* pad clearance in Table 9–1)

Clearance size associated with an unsupported hole is:

Clearance diameter = drill + manufacturing allowance in accordance with IPC-2221 + (2* pad clearance in Table 9–1)

Where breakout is allowed, increase clearance radially by the breakout allowance.

When nonfunctional lands are removed the only reduction in clearance size is the annular ring component, since, with no land present it does not apply.

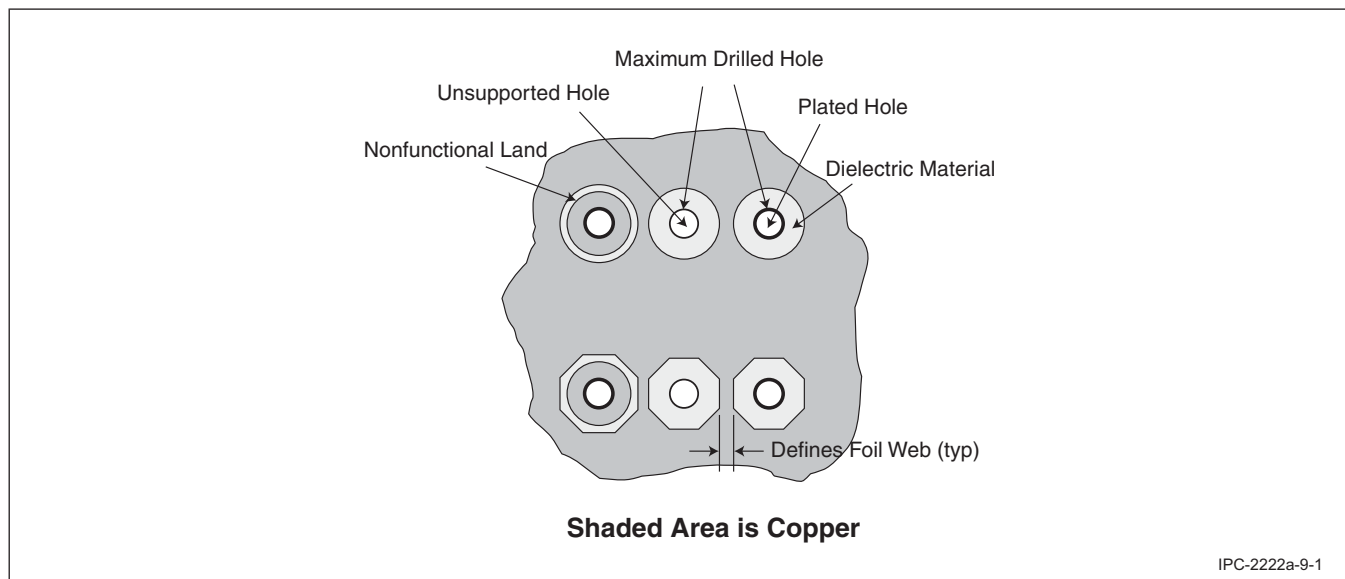


Figure 9–1 Clearance Area in Planes

Table 9–1 Pad to Plane Clearance, mm [in]

Level A	Level B	Level C
0.51 [0.020] or minimum electrical clearance, whichever is greater	0.25 [0.010] or minimum electrical clearance, whichever is greater	0.13 [0.005] or minimum electrical clearance, whichever is greater

9.1.3.1 Small Pitch Clearance Area in Planes A special precaution **shall** be observed when routing high-speed circuits and/or very small pitch devices. When routing small pitch devices and/or vias routed on small grids, designers **shall** remain aware of factors relating to power/ground plane clearance areas. When the pitch is very small, designers **shall** remain cognizant of the narrow foil web between clearance openings (see Figure 9–2). As the clearance area (diameter) is made larger, the foil web between clearance areas becomes smaller. Designs having very small foil webs are less desirable because of their reduced current carrying capability, potential for increased voltage drop, higher EMI emissions, and reduced thermal dissipating characteristics. It is highly desirable for heat generating devices to be “heatsinked” down through via holes and dissipated across the surface of inner planes. For these reasons the foil web should be as large as possible and over-lapping clearance areas in planes should be avoided. Use the following formula when dealing with small pitch devices and/or via holes, when routing on very small grids.

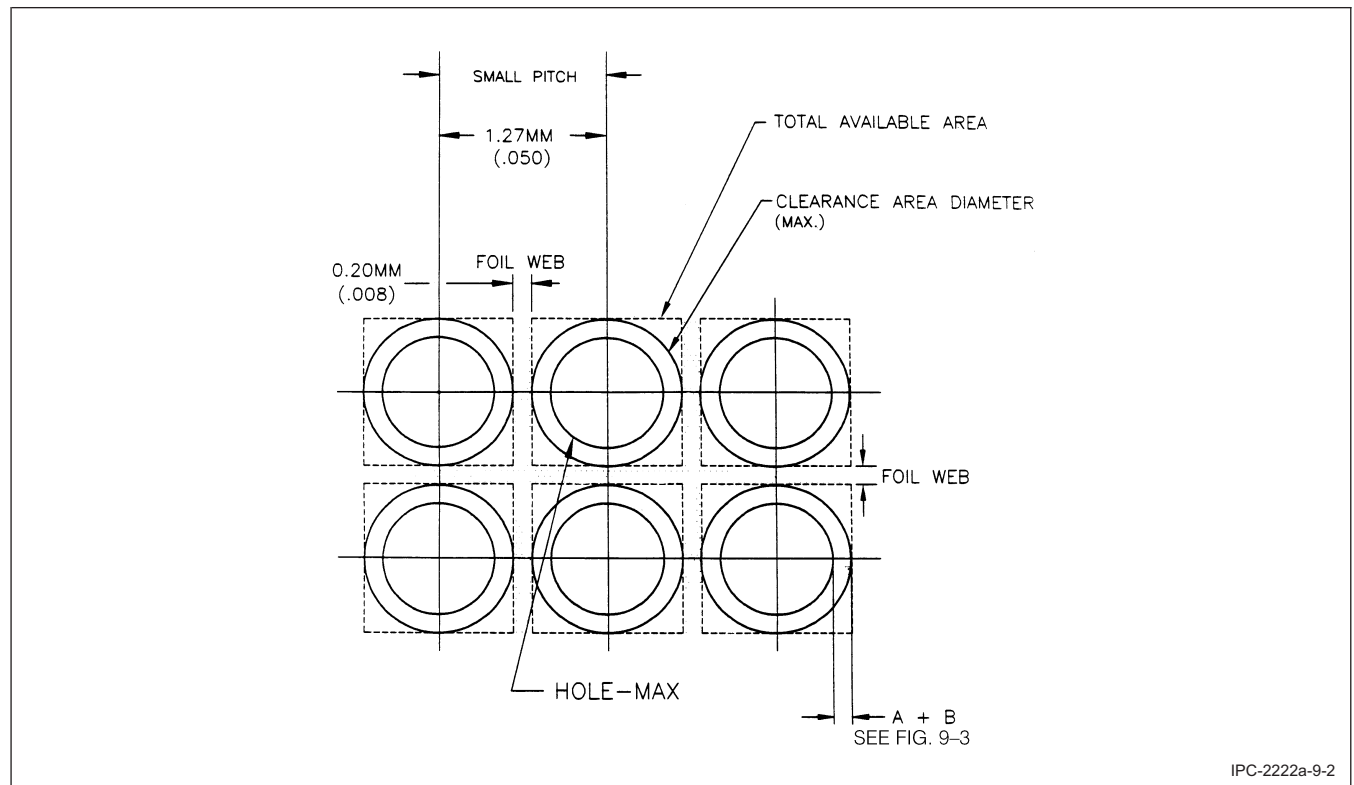


Figure 9-2 Foil Web Size

Typical example:

Determine the desired width of foil web.

Ex: 0.20 mm [0.0079 in] Web

$$\begin{aligned}
 \text{Total available area} &= \text{pitch} - \text{foil web} \\
 &= 1.25 - 0.20 \text{ mm} [0.0492 - 0.0079 \text{ in}] \\
 &= 1.05 \text{ mm} [0.0413 \text{ in}]
 \end{aligned}$$

$B = 0.25 \text{ mm} [0.0098 \text{ in}]$ min. fabrication allowance per Table 9-1 of IPC-2221.

$$\begin{aligned}
 \text{Clearance area (diameter)} &= \text{hole diameter (max.)} + 2B \\
 &= 0.35 \text{ mm} + 2 (0.25 \text{ mm}) [0.0138 \text{ in} + 2(0.0098 \text{ in})] \\
 &= 0.35 \text{ mm} + 0.50 \text{ mm} [0.0138 + 0.0197 \text{ in}] \\
 &= 0.85 \text{ mm} [0.0335 \text{ in}]
 \end{aligned}$$

Note: The maximum clearance area (diameter) in the plane is calculated by using the formula above and **shall** not exceed the total available area.

9.1.4 Nonfunctional Lands Nonfunctional lands are lands on internal layers that do not make an electrical connection. Nonfunctional lands should be included on internal layers for all PTHs except for the following design situations:

- In areas where there is sufficient electrical clearance but insufficient manufacturing clearance, typically such as on ground, voltage, or thermal planes.
- Where printed boards have greater than ten layers, it is allowable to remove nonfunctional lands in the vertical stack on alternate layers, preferably starting in the middle third of the printed board.

In these cases, the plated barrels **shall** meet the same minimum spacing requirements as that for conductors on internal layers. The design **shall** use the standard fabrication allowance to account for PTH position and artwork misregistration, even though nonfunctional lands are not present, because the barrel of the hole is a conductor.

PTHs passing through internal ground, voltage and thermal planes without nonfunctional lands **shall** meet the same minimum spacing requirements as conductors on internal layers. It is preferred that the space-to-PTH clearance not change because the lands are missing.

If nonfunctional lands are to be removed during the design process, it should be delayed until routing is completed in order to maximize conductor to copper barrel clearances.

Non-functional lands on plane layers are sometimes necessary to assure reliability. This is especially true when brittle resin systems like polyimide or high temperature epoxy are used, and is more likely with thick copper. On plane layers, the clearance surrounding the via or non-functional pad is completely filled with resin, since glass fibers cannot be pushed into these tight areas. These small resin rich areas may crack when exposed to drilling or thermal shock from hole processing. A non-functional land provides a copper barrier between the hole and the clearance area that will insulate this resin rich area from the effects of hole formation. Prepregs with filled resins may be available to mitigate or reduce the potential for cracking when nonfunctional lands are removed. The tradeoff between nonfunctional land removal and desired manufacturing clearance should be discussed with the printed board supplier.

Note: A fabricator **shall** not remove nonfunctional lands without specific approval of the design activity or unless allowed on the master drawing.

The value of non-functional lands is a complex and controversial subject and depend on the features of the individual design. Some of the concerns with non-functional lands are shown in Table 9-2.

Table 9-2 Nonfunctional Land Considerations

Reasons to Retain Non-functional Lands	Reasons to Remove Non-functional Lands
Omitted lands allow some routers to run conductors through the holes.	Each conductor in close proximity to a land is an area susceptible to potential etching process defects. The presence of more lands increases the potential for more defects.
Drawing review of interferences is easier when all holes by their lands can be visually identified	Lands can take up considerable space within a database.
Clearance is required for signal and plane layers for the maximum hole size, the misregistration, the manufacturing tolerance and the electrical spacing. This is the same clearance as that for functional lands with the exception of the annular ring requirement.	As copper is heavier than dielectric, boards utilizing fewer lands will be lighter.
Resin cracks are alleviated. Resin rich areas may crack when exposed to drilling or thermal shock from hole processing.	Some board manufacturers delete non-functional lands. If such lands are removed in the design, then the buyer can be assured that all of their hardware is in the same configuration.
There may be less propensity for innerlayer separation depending on z-axis expansion of the particular design.	There may be less propensity for barrel cracking depending on z-axis expansion of the particular design.
	The pad stack is thicker than the areas of the printed board without lands. Lamination pressure will tend to generate resin starvation and cracked glass fibers.
	Reduction of noise as stub lines.

9.1.5 Conductive Pattern Feature Location Tolerance The presentation in Table 9-3 is for the tolerance to be applied to the nominal dimension chosen for the location of the lands connector contacts and conductors in relation to the datums. This tolerance includes tolerances for master pattern accuracy, material movement, layer registration and fixturing.

Table 9-3 Feature Location Tolerances (Lands, Conductor Pattern, etc.) (Diameter True Position)¹, mm [in]

Greatest Printed Board/ X,Y Dimension	Level A	Level B	Level C
Up to 300 [11.8]	0.30 [0.0118]	0.20 [0.0079]	0.10 [0.0039]
Up to 450 [17.7]	0.35 [0.0138]	0.25 [0.0098]	0.15 [0.0060]
Up to 600 [23.6]	0.40 [0.0157]	0.30 [0.0118]	0.20 [0.0079]

Note 1. Conductor pattern registration may be expressed in terms of minimum annular ring violation, which establishes manufacturing registration allowances.

9.2 Holes Holes **shall** be in accordance with the generic standard IPC-2221 and as follows:

9.2.1 Unsupported Holes

9.2.1.1 Unsupported Hole Diameter One of the primary applications for unsupported holes is to provide clearance for mounting hardware. This hardware may be critical in aligning the electronic assembly to other assemblies or housings. When selecting a hole size and tolerance, the designer should perform a tolerance analysis that takes into consideration the diameter tolerance for the mounting hardware and unsupported hole positional tolerance required to insure alignment to the next assembly. The nominal unsupported hole size should be a readily available drill size.

When using the basic dimensioning system, holes **shall** be expressed in terms of maximum material condition (MMC) and least material condition (LMC) limits. The tolerance ranges shown in Table 9-4 are used to determine the MMC-LMC limits.

Table 9-4 Diameter Tolerance Range for Unsupported Holes, mm [in]

Level A	Level B	Level C
0.20 [0.008]	0.15 [0.006]	0.10 [0.004]

9.2.1.2 Eyelet Hole Diameter When eyelets are used, the diameter of holes in which eyelets are inserted **shall** not exceed the outside diameter of the barrel of the eyelet by more than 0.15 mm [0.0060 in]. The relationships between maximum and minimum barrel diameters and wire diameters **shall** be as shown in Table 9-5.

Table 9-5 PTH Diameter to Lead Diameter Relationships, mm [in]

Lead Diameter	Level A	Level B	Level C
Minimum hole to minimum lead diameter	No greater than 0.7 [0.028] over minimum lead diameter	No greater than 0.7 [0.028] over minimum lead diameter	No greater than 0.6 [0.024] over minimum lead diameter
Minimum hole to maximum lead diameter	No less than 0.25 [0.0098] over maximum lead diameter	No less than 0.20 [0.0079] over maximum lead diameter	No less than 0.15 [0.006] over maximum lead diameter

9.2.2 Plated-Through Holes (PTHs) The maximum and minimum PTH diameters used to attach component leads or pins to the printed board **shall** be evaluated in accordance with Table 9-5. Both minimum and maximum leads **shall** be taken into consideration in evaluating the finished PTH requirements. If the lead is a ribbon lead, the minimum and maximum diagonal of a flat ribbon lead **shall** be considered. Table 9-5 shows the limits of the PTH.

These limits **shall** be optimized so that manufacturability is enhanced to provide the most liberal tolerances allowable (see Figure 9-3).

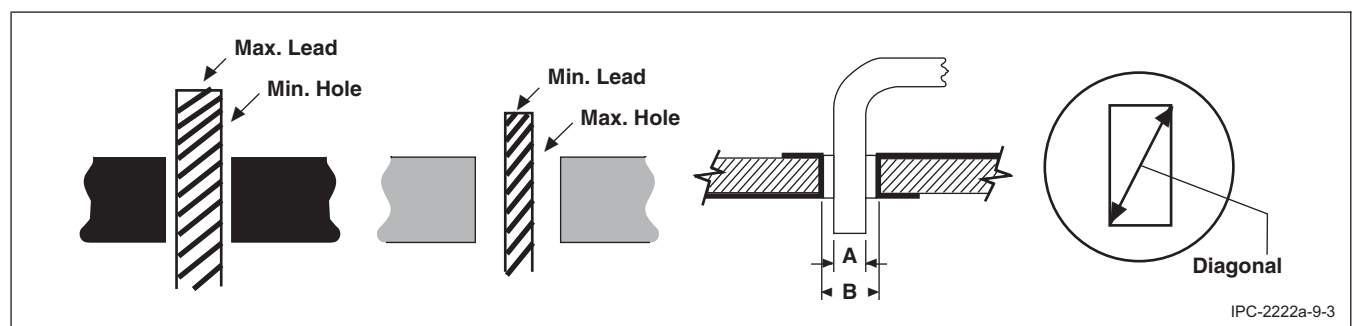


Figure 9-3 Lead-to-hole Clearance

Unless otherwise specified, the hole size **shall** be the finished plated size after solder coating or final plating and fusing, if required. The hole size **shall** be specified on the master drawing. PTHs used for functional interfacial connections **shall** not be used for the mounting of devices, which put the PTH in compression. PTHs used for functional interfacial connections **shall** not be used for the mounting of eyelets, solder terminals, or rivets. PTHs **shall** be used for all interfacial connections on multilayer printed boards Type 3 through Type 6 (inclusive). Platings and coatings **shall** be in accordance with IPC-2221.

9.2.2.1 Aspect Ratio Aspect ratio is the ratio of the nominal printed board thickness (over external foils) to the maximum drilled hole diameter of the smallest PTH. As the printed board thickness increases, so does the probability of drill bit wander during drilling operations. As the PTH size decreases, it becomes more difficult to expose the barrel of the PTH to the plating

solutions. The aspect ratio of PTHs plays an important part in the ability of the manufacturer to provide sufficient plating within the PTH, as well as in the reliability of the PTH/PTV structure (see IPC-D-279). Table 9–6 provides information on the producibility of aspect ratios for various levels of complexity. Aspect ratios 9:1 or greater are not recommended for hole sizes less than 0.25 mm [0.010 in] diameter.

Table 9–6 PTH Aspect Ratio

	Level A	Level B	Level C
Aspect Ratios	< 6:1	6:1 to 8:1	> 8:1

9.2.2.2 Specifying Hole Sizes for Component Leads When using the basic dimensioning system, PTHs used to attach component leads or pins to the printed board should be expressed in terms of MMC and LMC limits. The tolerances shown in Table 9-7 are used to determine the MMC-LMC limit for the appropriate finished hole diameter. Thus, a hole 1.0 ± 0.05 mm [0.040 ± 0.002 in] would be expressed as $0.95 - 1.05$ mm [$0.038 - 0.042$ in].

Table 9-7 Minimum PTH Diameter Tolerance Range, (Difference Between High and Low Hole Size Limits), mm [in]

Level A	Level B	Level C
0.20 [0.0079]	0.15 [0.0060]	0.10 [0.0040]

9.2.2.3 Specifying Hole Sizes for Vias The finished via size should be represented on the master drawing as the maximum via dimension. No minimum via dimension should be specified, since the via contains no component lead or pin and could theoretically be plated shut or closed after final finishing.

When via life in harsh environments is a consideration, the minimum drilled via size should be controlled to maintain an aspect ratio that is known to be reliable. Allowing vias to be plated shut solves a producibility concern; however, this allows the supplier to use a smaller drill to improve registration, which would increase the aspect ratio. A better approach for high reliability products is to specify the minimum drill diameter to control the aspect ratio.

9.2.3 Etchback Etchback, when required, will reduce the annular ring support on internal layers of the printed board. Therefore, this should be taken into account when specifying PTH land size. However, the maximum etchback allowed on the master drawing **shall** not be greater than the minimum design annular ring.

10 GENERAL CIRCUIT FEATURE REQUIREMENTS

10.1 Conductor Characteristics Conductor characteristics **shall** be in accordance with the generic standard IPC-2221 and as follows:

10.1.1 Edge Spacing Except for edge printed board contacts, the minimum distance between conductive surfaces and the edge of the finished printed board, or a NPTH, **shall** not be less than the minimum spacing specified in Table 6–1 of IPC-2221 plus 0.4 mm [0.0157]. Printed boards that slide into guides **shall** have a minimum external conductor to guide distance of 1.25 mm [0.0492 in] or minimum electrical clearance (see Table 6–1 of IPC-2221), whichever is greater. Special design applications in areas such as high voltage, surface mount, and radio frequency (RF) technology may require variances to these requirements. Ground and heat sink planes may extend to the edge when required by design.

The design should provide sufficient edge spacing to avoid unnecessary haloing and crazing rejections. Rejections can occur when the edge defect is greater than 50% of the clearance to edge spacing. Brittle material or heavier glass weights increase this risk.

10.1.2 Balanced Conductors Whenever possible, to reduce bow and twist and to increase dimensional stability, conductors should be balanced within an individual layer. Conductor routing density should be spread throughout the printed board wherever possible, to avoid the need for special etching or plating thieves.

Plating thieves are added metallic areas which are nonfunctional within the finished printed board profile but allow uniform plating density, giving uniform plating thickness over the printed board surface.

The multilayer printed board structure should also be as balanced as possible providing equal layers of signal conductors and planes to either side of the center of the multilayer construction.

10.1.3 Flush Conductors for Rotating or Sliding Contacts When flush circuits are required for application as mating contact surfaces, the degree of flushness **shall** meet the requirements of Table 10–1. Figure 10–1 and Figure 10–2 show a typical flush circuit design.

Note: The level of flushness required is normally a function of factors related to the mating contact, such as contact size, shape, load, rotational force, surface friction, etc. It is recommended that the designer fully understand the dynamics of the mating contact system before establishing surface flushness requirements.

Table 10–1 Surface Flushness Requirements, mm [in]

Level A	Level B	Level C
AABUS	± 0.013 [0.000512]	± 0.005 [0.000197]

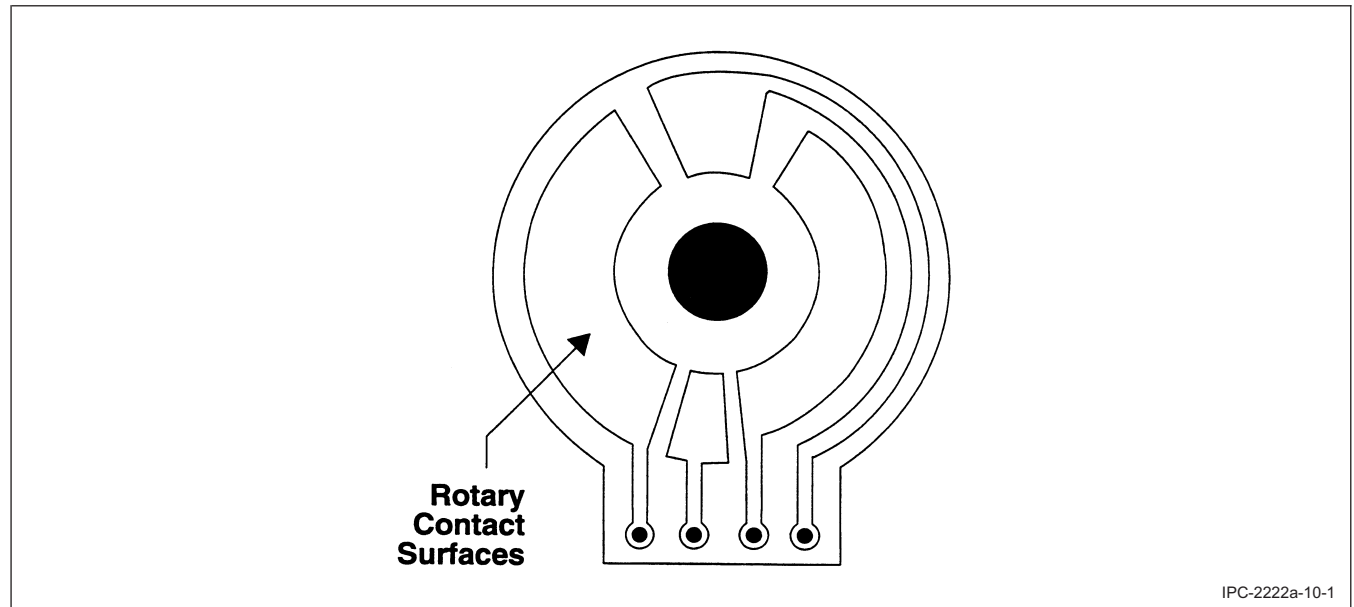


Figure 10–1 Typical Flush Circuit

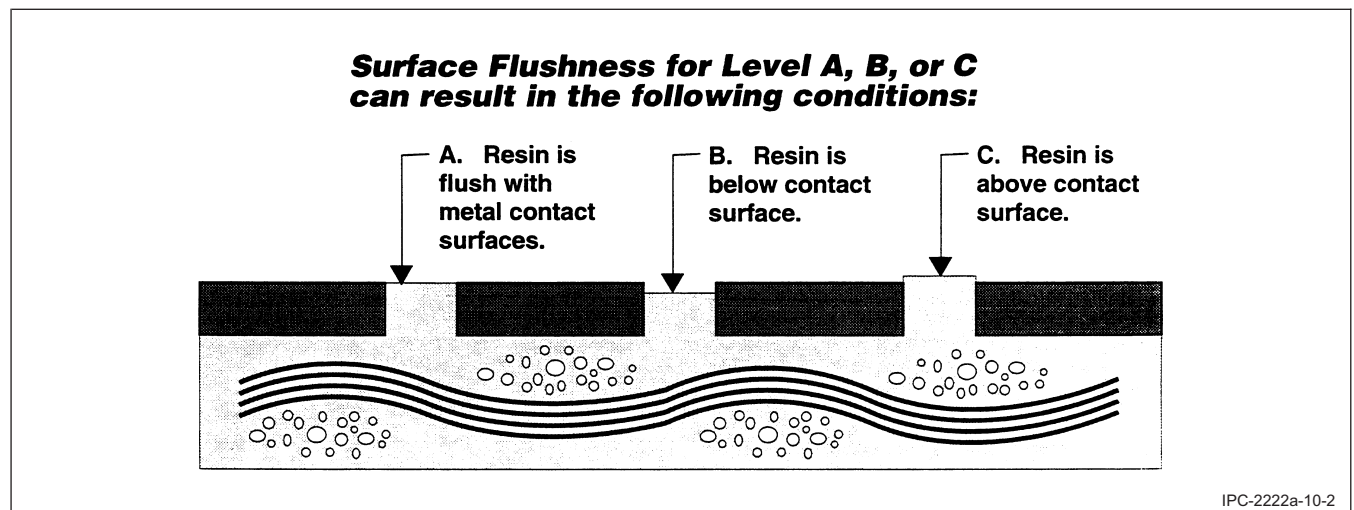


Figure 10–2 Surface Flushness Conditions

10.1.3.1 Metallic Finishes for Flush Conductors Metallic finishes for flush conductor contacts should be gold over nickel, or other suitable corrosion resistant, low contact resistance finish.

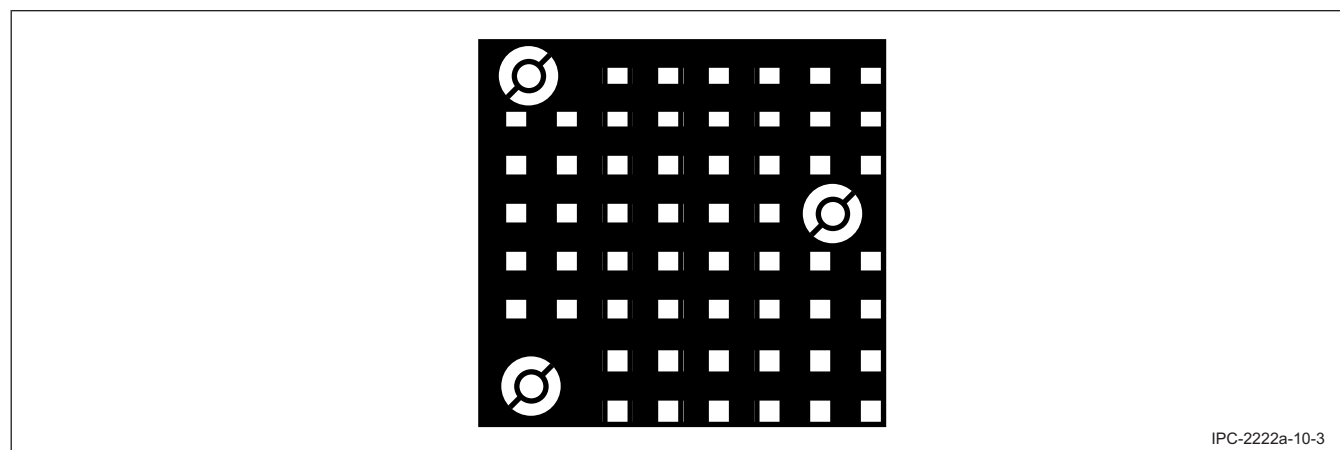
10.2 Land Characteristics Land characteristics **shall** be in accordance with the generic standard IPC-2221 and as follows in 10.2.1. Lands for interfacial connections **shall** meet the requirements of 10.1.

10.2.1 Offset Lands Lands, when used in conjunction with clinched leads, may be located adjacent to (not surrounding) the lead termination hole. The land **shall** be a sufficient distance from the hole to allow clipping of the part lead prior to unsoldering the part lead from the land.

10.3 Large Conductive Areas Large conductive areas (planes 3.0 mm [0.118 in] wide or larger conductors) increase the likelihood for blistering, warping, or heat shielding during wave or reflow soldering.

Large areas that cover more than a 25.0 mm [0.984 in] diameter may be broken up into a cross-hatched or similarly patterned area (see Figure 10–3). Modifications to large conductive areas **shall not** adversely impact the electrical characteristics or performance of the printed board. For designs intended for wave soldering, large conductive areas should not be on the solder side of the printed board unless protected by solder mask or other method.

When UL requirements are imposed, the area **shall** be within the limits approved by UL for the printed board manufacturer. UL testing of conductive patterns for blistering, warping and delamination issues is determined by a maximum, non-pierced conductive area defined by the largest circle that can be inscribed within a conductive pattern. This conductive area, referred to as the maximum area diameter (MAD), is listed on UL printed board cards. The usage of cross-hatched patterns as shown in Figure 10–3 therefore allows for large conductive area diameters, as UL tests only the non-pierced conductive areas.



IPC-2222a-10-3

Figure 10–3 Cross-hatched Large Conductive Layers with Isothermal Conductors

11 DOCUMENTATION

Documentation **shall** be in accordance with the generic standard IPC-2221 and as follows:

11.1 Filled Holes Via holes designated to be filled by the design requirements **shall** be identified on the drawing.

11.2 Nonfunctional Holes Electrically nonfunctional supported holes should be identified as such on the fabrication or assembly drawing and do not need to meet the electrical connection requirement.

12 QUALITY ASSURANCE

Quality Assurance **shall** be in accordance with the generic standard IPC-2221.



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